SIGMARHO KALMAN FILTER IMPLEMENTATION

"Kalman Filter Implementation with Improved Numerical Properties"

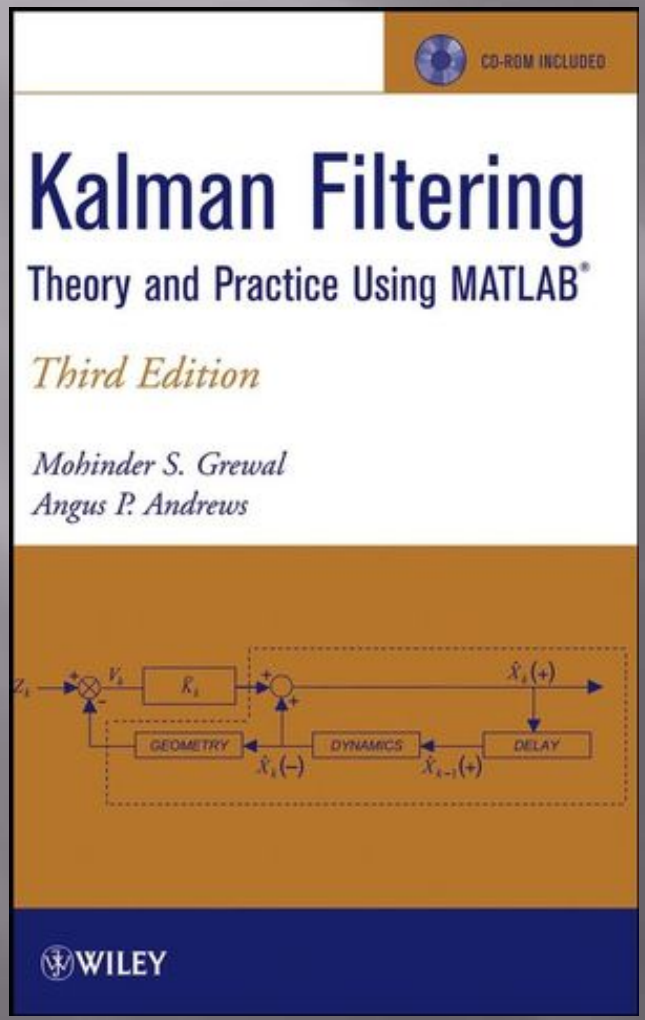
Accepted for publication in IEEE Transactions on Automatic Control

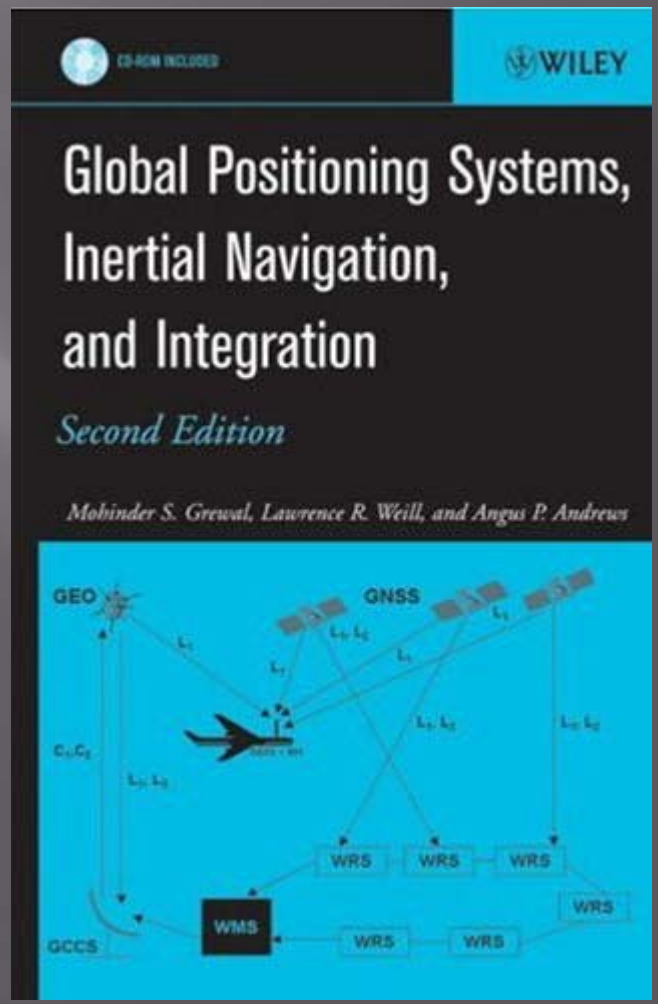
Dr Mohinder Grewal and James E. Kain

NUMBER FP-09-062.1

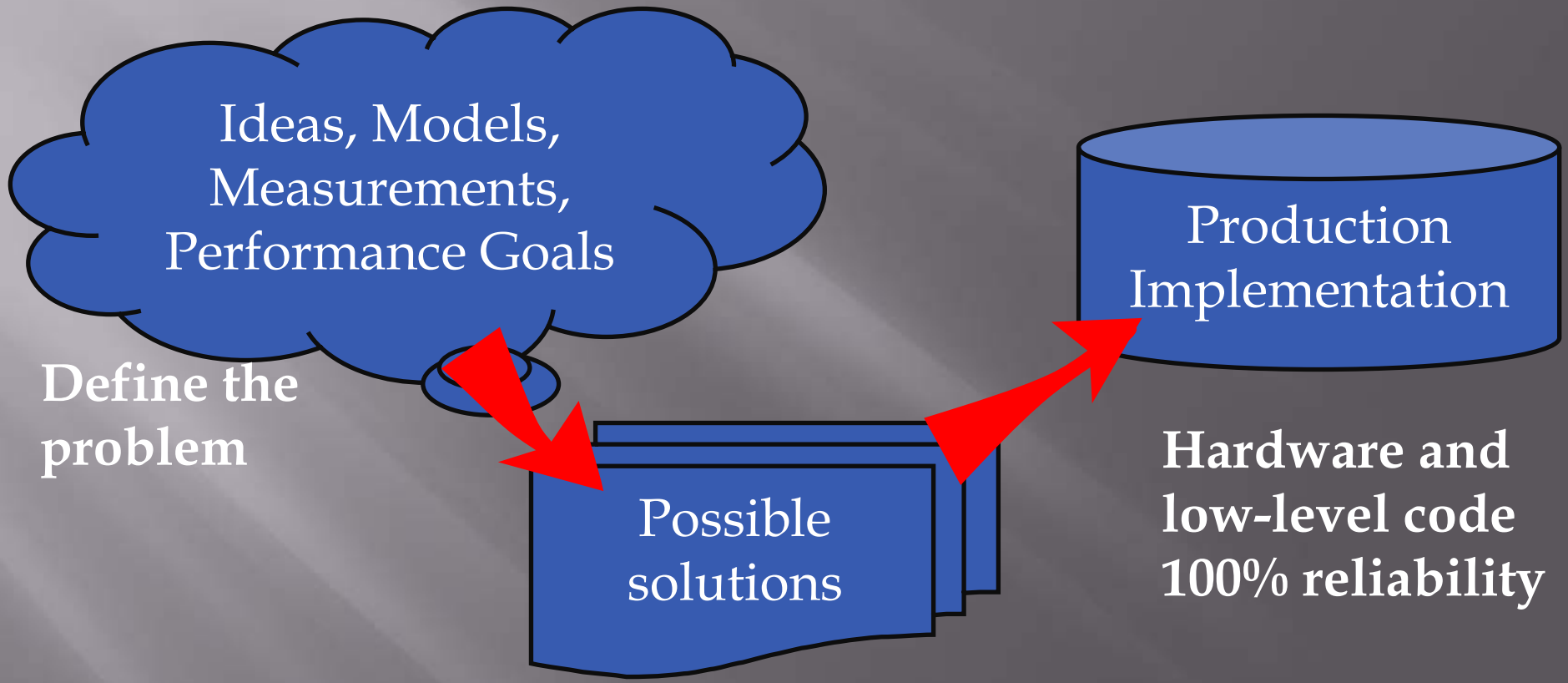
Dr Mohinder Grewal

Professor of Electrical Engineering California State University, Fullerton where he received the 2009 Outstanding Professor award.





From Engineering Concepts to Production Implementation

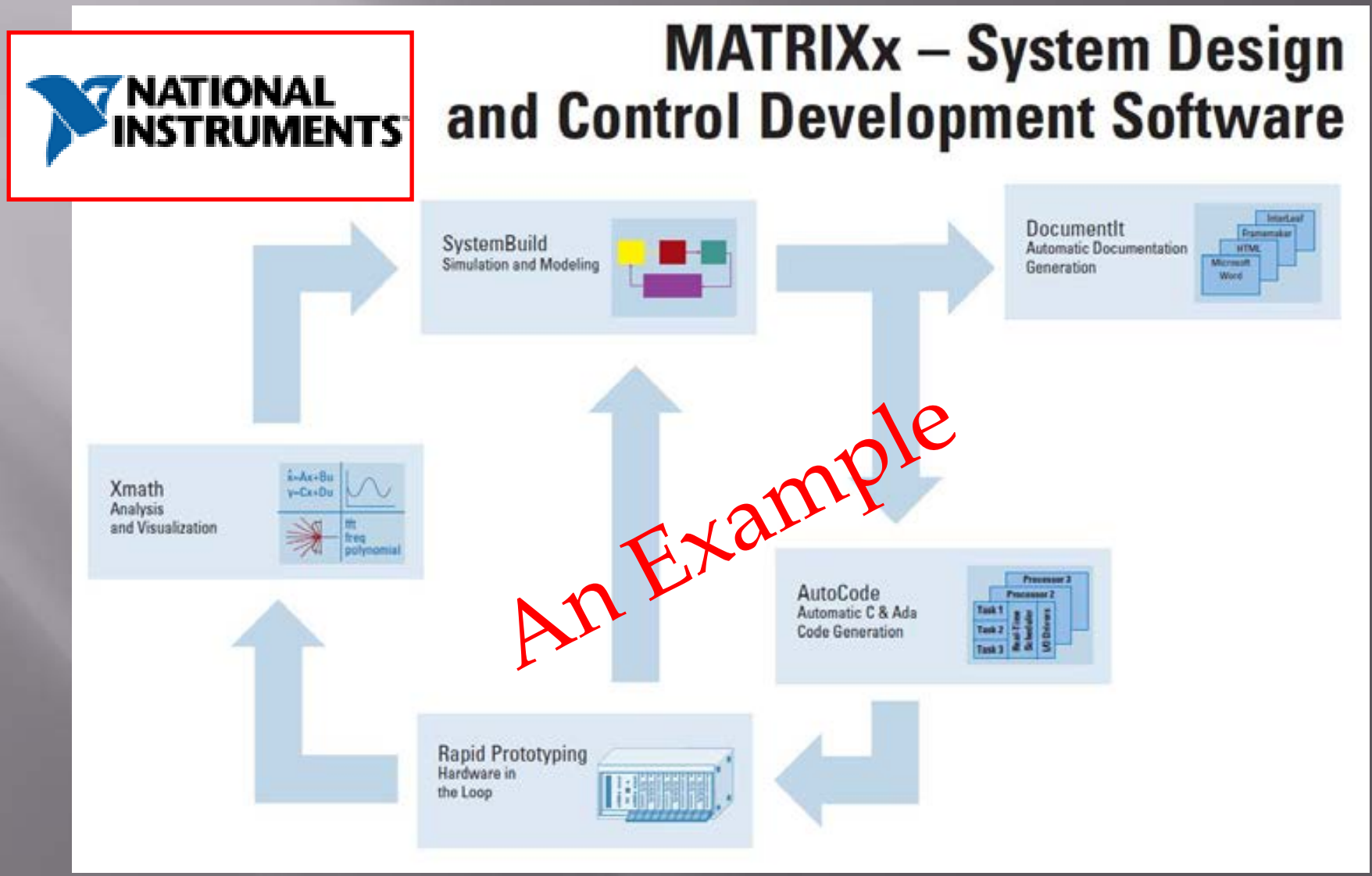


Define the problem

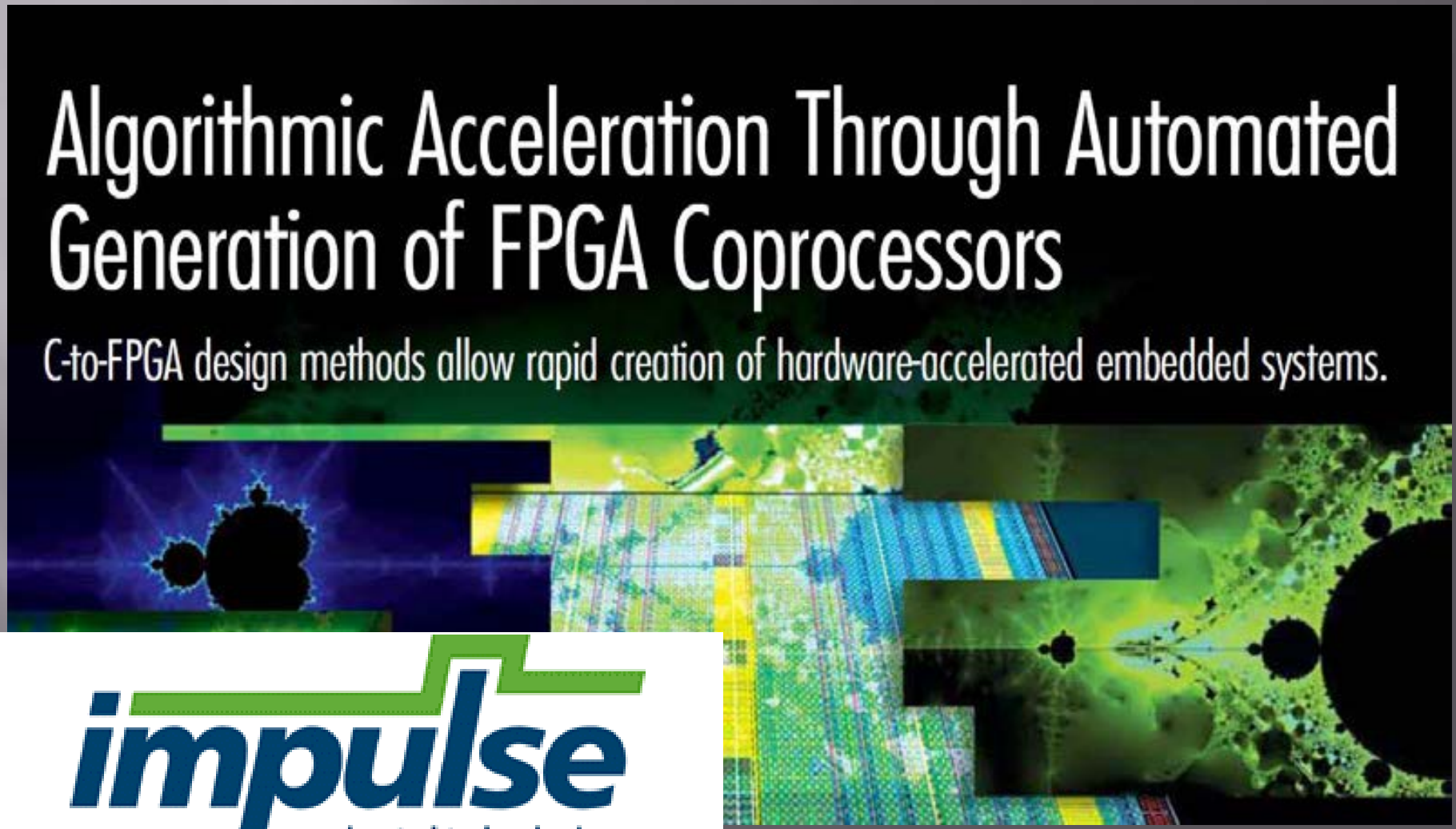
Hardware and low-level code 100% reliability

Simulate, implement & test with high-level code

Closing the Design>Implementation Gap



Closing the Design>Implementation Gap



Common Kernels as Building Blocks for all Algorithms

View From Berkeley

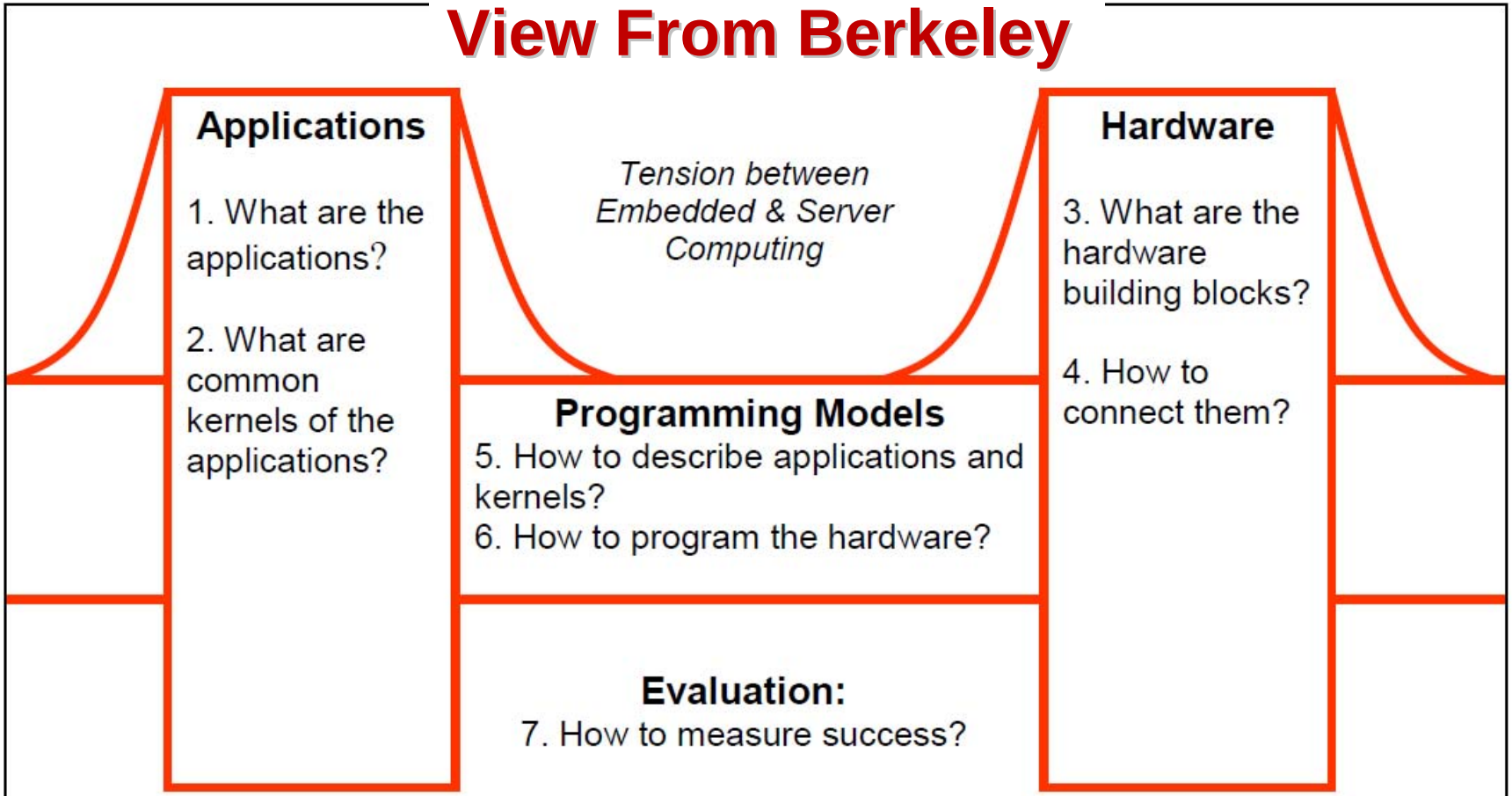


Figure 1. A view from Berkeley: seven critical questions for 21st Century parallel computing. (This figure is inspired by a view of the Golden Gate Bridge from Berkeley.)

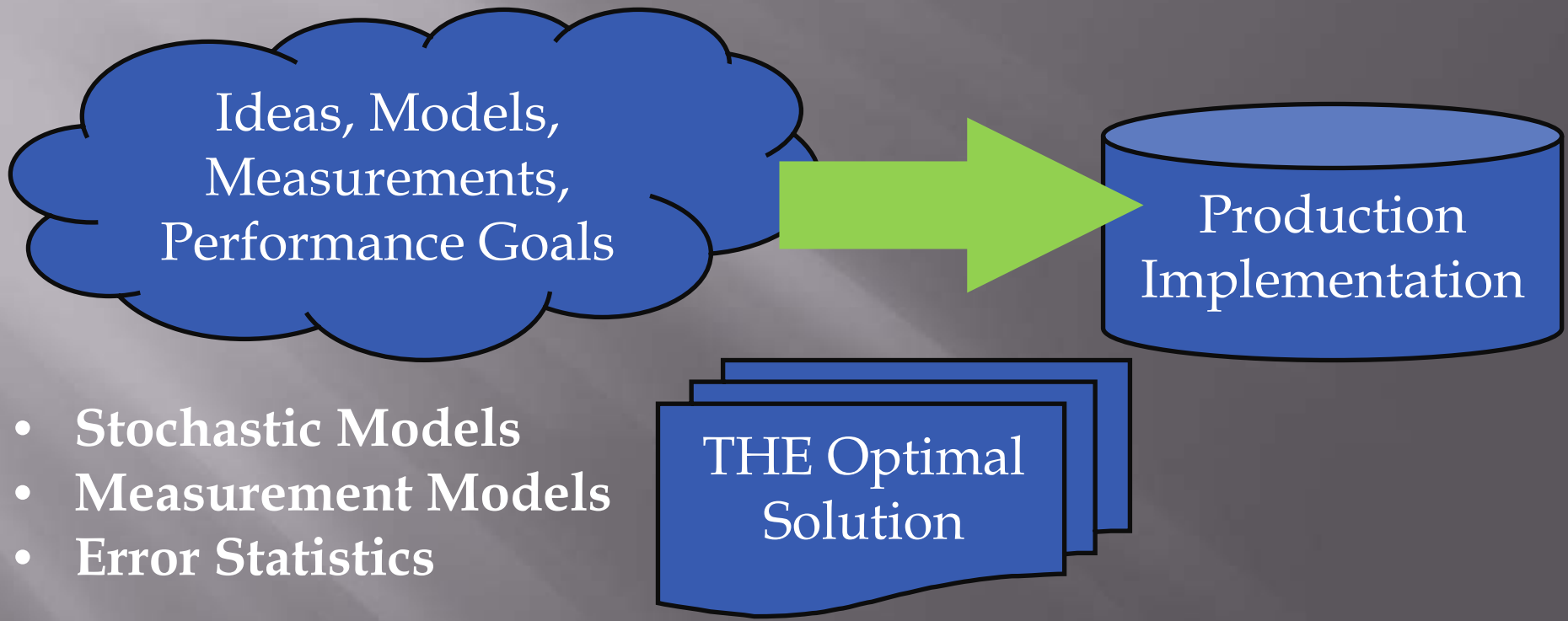
Seven Dwarfs

“The dwarfs present a method for capturing the common requirements of classes of applications while being reasonably divorced from individual implementations.”

1. **Dense Linear Algebra**
2. **Sparse Linear Algebra**
3. **Spectral methods**
4. **N-Body Methods**
5. **Structured grids**
6. **Unstructured grids**
7. **Monte Carlo Methods**

[Colella 2004] P. Colella, “Defining Software Requirements for Scientific Computing,” presentation, 2004.

1970's Promise of Modern System Theory

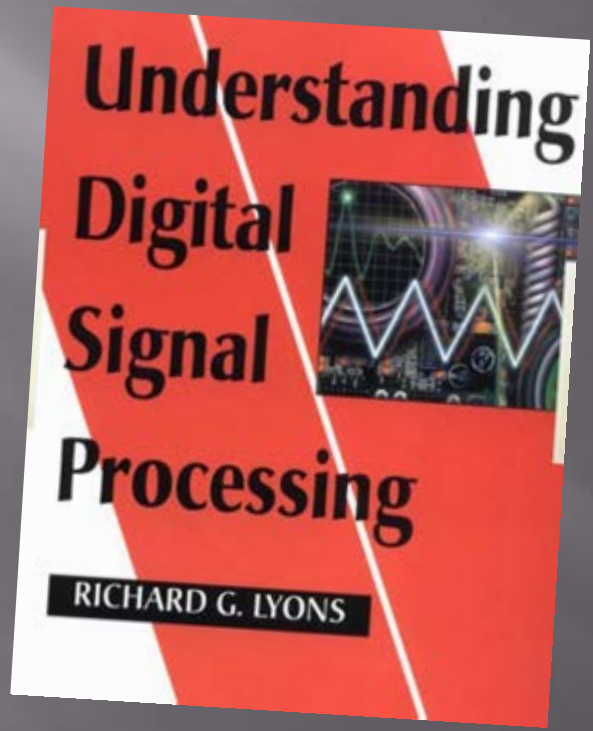


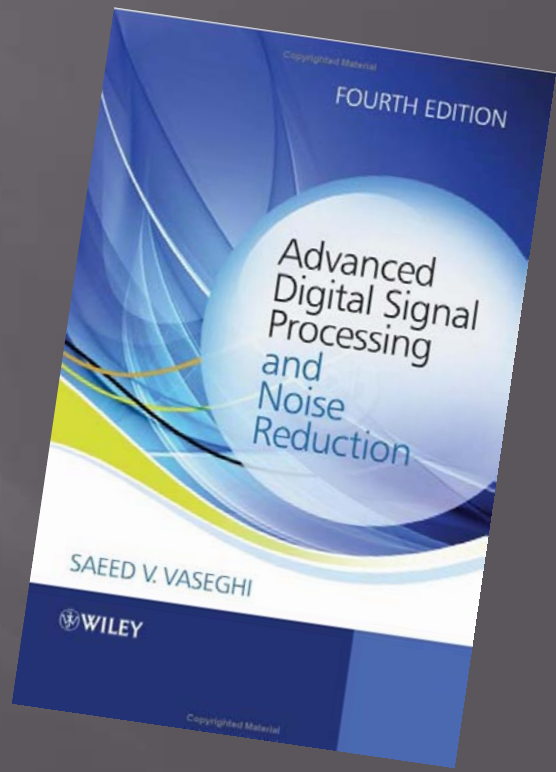
- **Stochastic Models**
- **Measurement Models**
- **Error Statistics**

Accelerate the Engineering Process!

Why Isn't the Kalman Filter used for Signal Processing?







Possible Reasons…

- **After 30 years of implementations, robustness concerns remain**
  - ✓ Square root filtering
  - ✓ Fictitious noise injection
  - ✓ Fading memory filters
  - ✓ Initialization issues
  - ✓ Residual tracking/adaptation
  - ✓ "Tuning"
- **No straightforward path to fixed point**

Sigma Rho Filter

- Propagate standard deviation (σ) and correlation coefficients (ρ) rather than covariance matrix or square root factors
  - ✓ Square root-like behavior
  - ✓ Natural scaling options
  - ✓ Physical interpretation
- Sigma and Rho trends provide information on current and projected robustness status
  - ✓ Rate of convergence
  - ✓ Limit of convergence
  - ✓ Correlation properties

Continuous System Propagation

Auxiliary variables	$\sigma'_i = \frac{\sigma_i}{\sigma_{Max_i}} \qquad F'_{ij} = \frac{F_{ik}\sigma_{Max_k}}{\sigma_{Max_i}} \qquad \mathrm{Q}'_{ij} = \frac{\mathrm{Q}_{ij}}{\sigma_{Max_i}\sigma_{Max_j}}$
Standard Deviation Propagation	$\frac{\dot{\sigma}'_i}{\sigma'_i} = \mathrm{M}'_{ii} + \frac{\mathrm{Q}'_{ii}}{2\sigma'^2_{ii}} \qquad \mathrm{M}'_{ij} = \frac{\sum_{k=1}^{n} F'_{ik}\sigma'_k\rho_{kj}}{\sigma'_i}$
Correlation Coefficient Propagation	$\dot{\rho}_{ij} = -\left(\frac{\dot{\sigma}'_i}{\sigma'_i} + \frac{\dot{\sigma}'_j}{\sigma'_j}\right)\rho_{ij} + \mathrm{M}'_{ij} + \mathrm{M}'_{ji} + \frac{\mathrm{Q}'_{ij}}{\sigma'_i\sigma'_j}$
State Propagation	$\dot{\underline{x}}'_i = \frac{f(\underline{x})}{\sigma_{Max_i}\sigma'_i} - x'_i\frac{\sigma'_i}{\sigma'_i} \;\; ; \;\; x_k = x'_k\sigma_{MAX_k}\sigma'_k$

Discrete System Propagation

Normalized Dynamics	$x_i' = \frac{\lambda x_i}{\sigma_{Max_j} \sigma_i'^{+}} ; \Phi_{ij}' = \frac{\Phi_{ij} \sigma_{Max_j} \sigma_j'}{\sigma_{Max_i} \sigma_i'} \; ; \; Q_{k_{ij}}' = \frac{Q_{k_{ij}}}{\sigma_{Max_i} \sigma_{Max_j} \sigma_i' \sigma_j'}$
Standard Deviation Propagation	$\left(\frac{\sigma_i'^{-}}{\sigma_i'^{+}} \right) = \sqrt{\left[\Phi' \Gamma \Phi'^{T} + Q_k' \right]_{ii}} \quad \Gamma_{ij} = \rho_{ij}^{+}$
Correlation Coefficient Propagation	$\rho_{ij}^{-} = \left(\frac{\sigma_i'^{+}}{\sigma_i'^{-}} \right) \left(\frac{\sigma_j'^{+}}{\sigma_j'^{-}} \right) \left[\Phi' \Gamma \Phi'^{T} + Q_k' \right]_{ij}$
State Propagation	$x_i'^{-}(k+1) = \left(\frac{\sigma_i'^{+}}{\sigma_i'^{-}} \right) \left[\Phi' x'(k) \right]_i$

Measurement Update

Auxiliary Parameters	$D_i = \sum_{k=1}^{n} H'_k \sigma'^{-}_k \rho^{-}_{ki} \;\; ; \;\; H'_i = H_i \sigma_{Max_i}$
Residual Variance	$\Omega^2 = \sum_{k=1}^{n} D_k \sigma'^{-}_k H'_k + R$
Standard Deviation Update	$\frac{\sigma'^{+}_i}{\sigma'^{-}_i} = \sqrt{1 - \left[\frac{D_i}{\Omega}\right]^2}$
Correlation Coefficient Update	$\rho^{+}_{ij} = \left[\frac{\sigma'^{-}_i}{\sigma'^{+}_i}\right]\left[\frac{\sigma'^{-}_j}{\sigma'^{+}_j}\right]\left[\rho^{-}_{ij} - \frac{D_i D_j}{\Omega^2}\right]$
State Update	$x'^{+}_i = \left[\frac{\sigma^{-}_i}{\sigma^{+}_i}\right]\left\{x'^{-}_i + \frac{D_i}{\Omega}\left[\frac{\lambda z_m - H'_k \sigma'_k x'_k}{\Omega}\right]\right\}$

Filter Adaptation

Measurement Noise Adaptation

Limit State improvement at a measurement

Test $\frac{\sigma_i'^{+}}{\sigma_i'^{-}} = \sqrt{1-\left[\frac{D_i}{\Omega}\right]^2}$ Control

Process Noise Adaptation (surgical control)

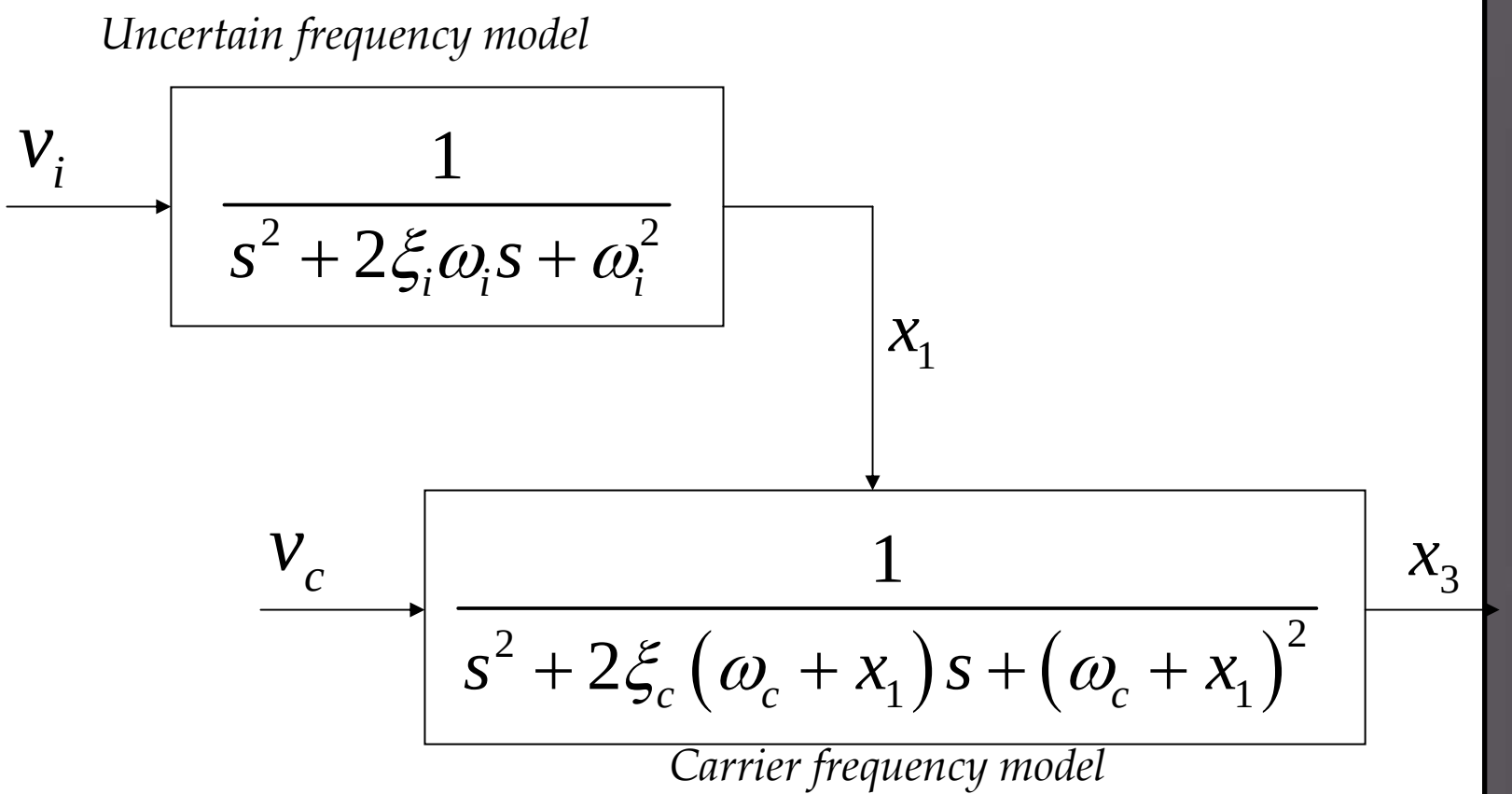
Limit correlation coefficients and lower sigma limit

Computed

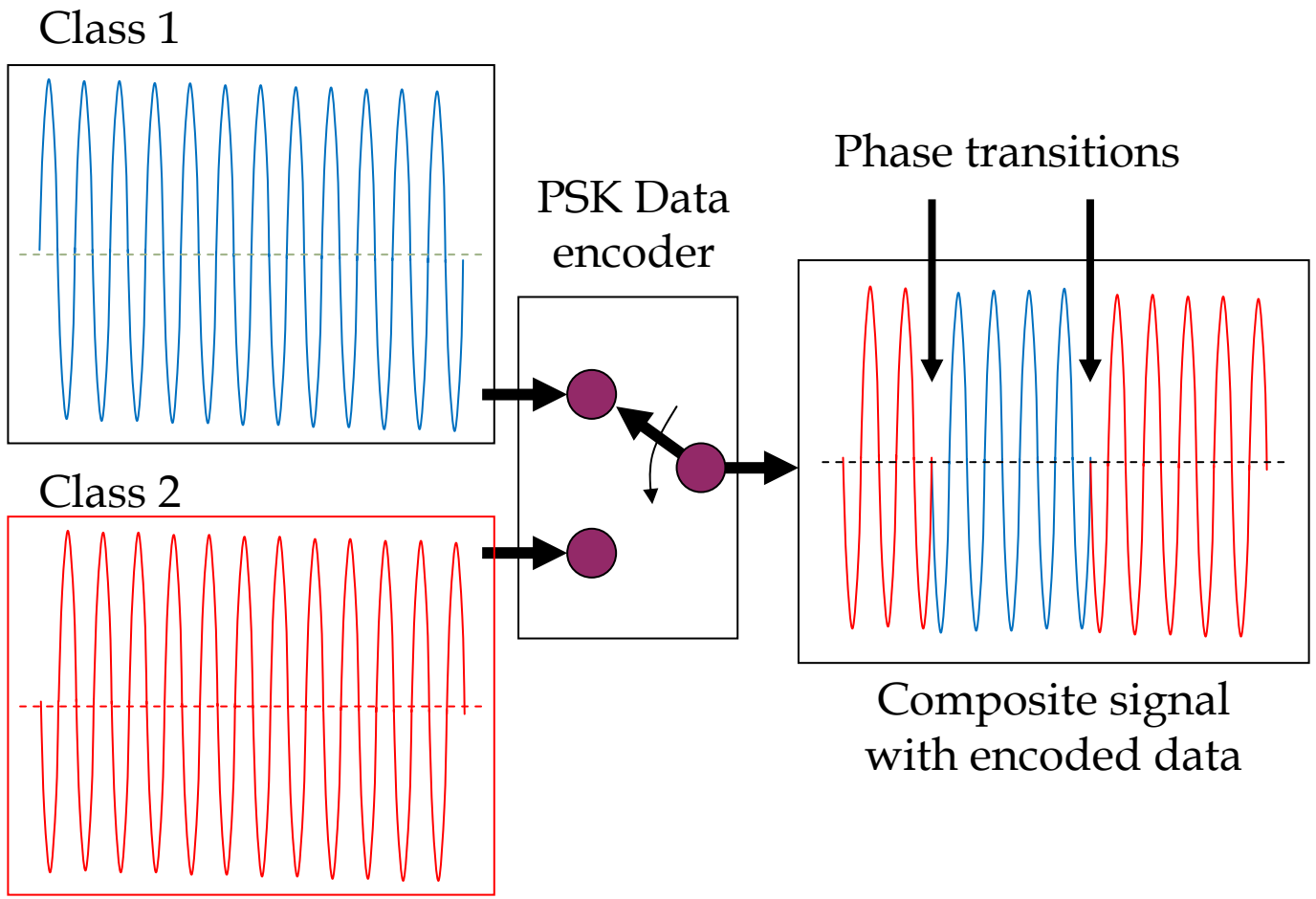
$$\rho_{ij}' = \frac{\rho_{ij}}{1+\gamma} \qquad \sigma_i'^{2} = \sigma_i^2\left(1+\gamma\right)$$

Adapted

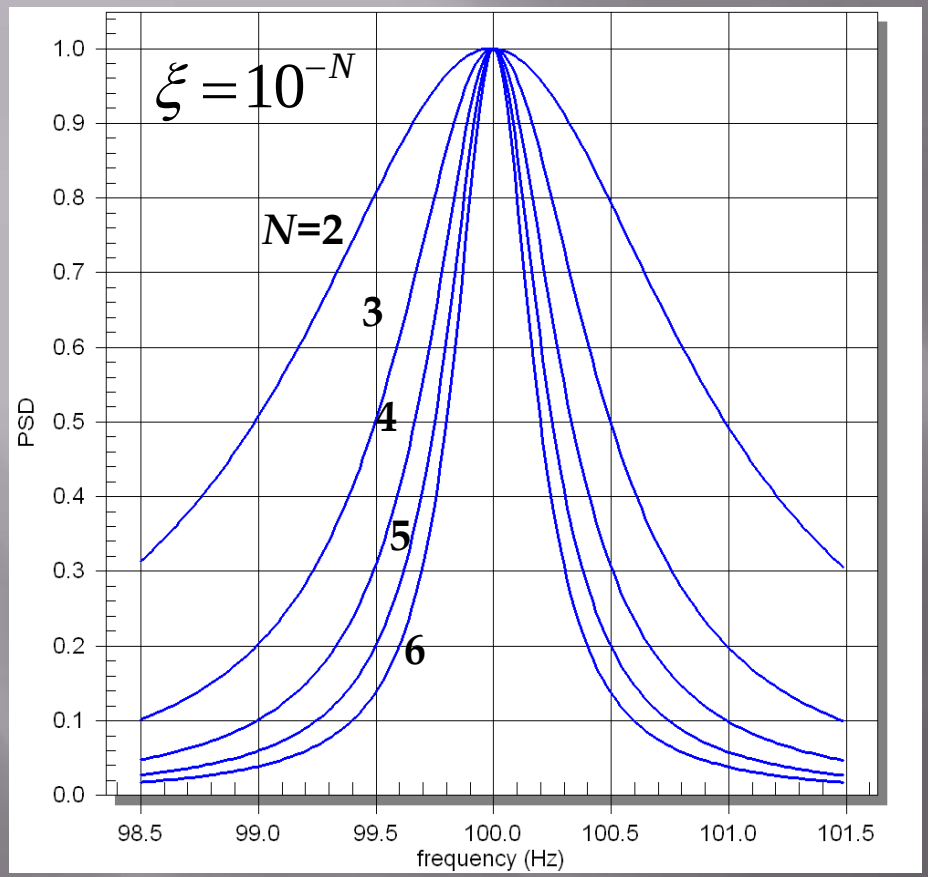
Sample Problem
Software Defined Radio



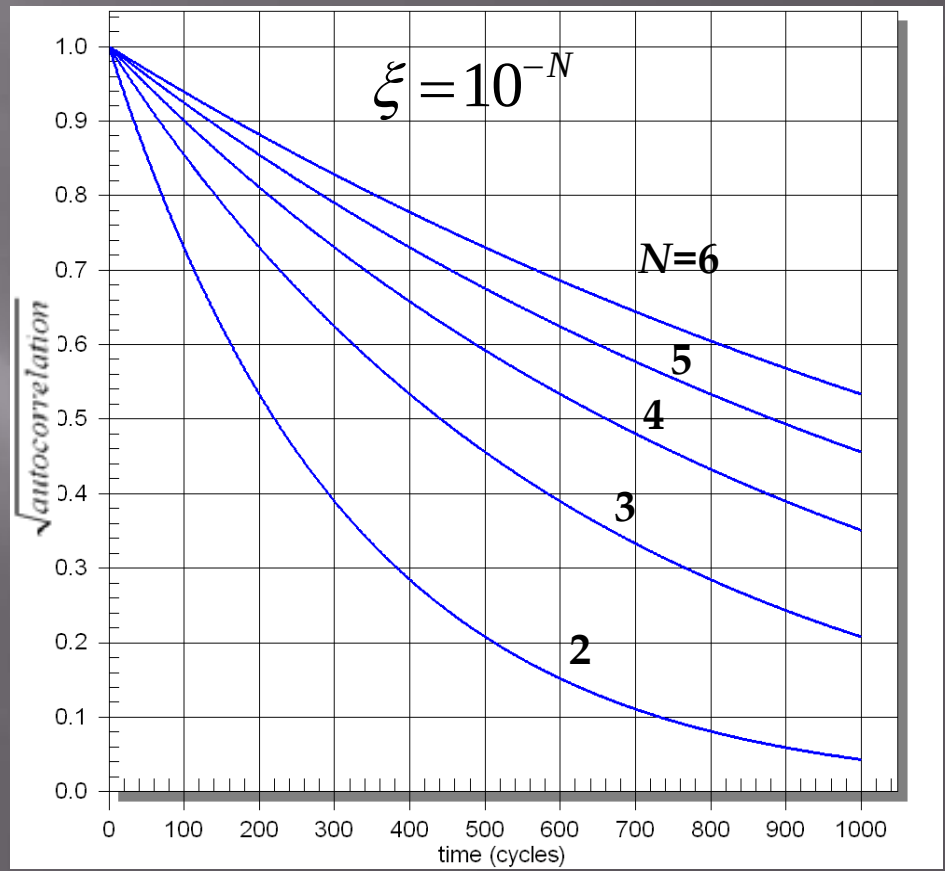
BPSK Encoding onto Carrier



Carrier Model Design



(a) Spectral response



(b) Autocorrelation

State Dynamics Equation

Significant state nonlinearity!

$$\begin{bmatrix} \dot{x}_0 \\ \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -2\xi_i\omega_i & -\omega_i^2 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -2\xi_c(\omega_c + x_1) & -(\omega_c + x_1)^2 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} w_i \\ 0 \\ w_c \\ 0 \end{bmatrix}$$

Fortunately!

Transition Matrix and Discrete Noise matrix can be solved in closed form with high accuracy

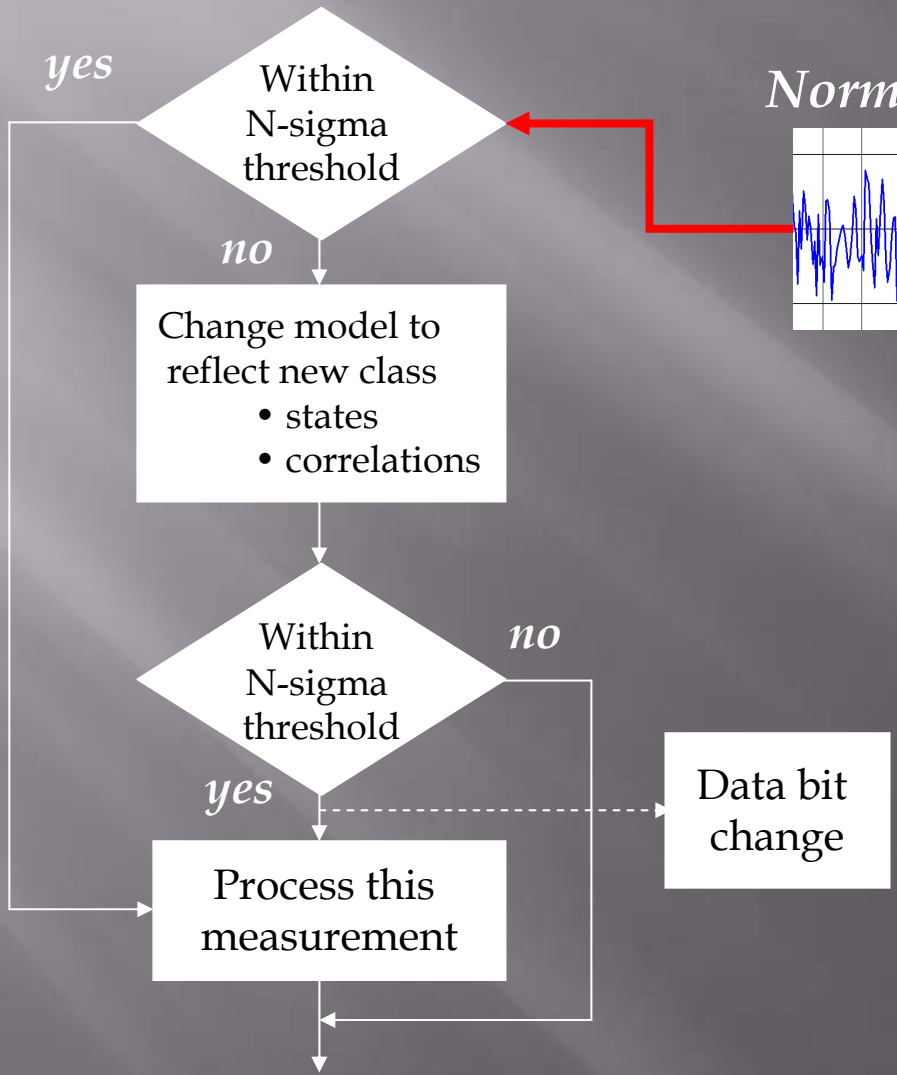
Assumptions:

- Information 2nd order model not influenced by carrier model
- Information bandwidth << carrier frequency

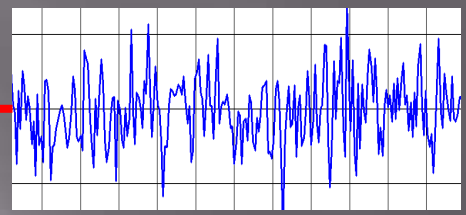
Model Parameters

parameter	Value
Carrier frequency	1.5 GHz
Carrier damping	10^{-8}
Information bandwidth	1.0 Hz
Information model damping	0.50
Sample rate	74MHz (1/20.5 carrier)
Encoding rate	7.5MHz (1/200 carrier)
Noise	10% carrier amplitude
Initial carrier frequency error	50 PPM
Carrier frequency error sigma	100 PPM
Information gain limit	50% / measurement
Sigma lower limit	10% initial value
Correlation limit	+/- 0.95

BPSK Data Decoding

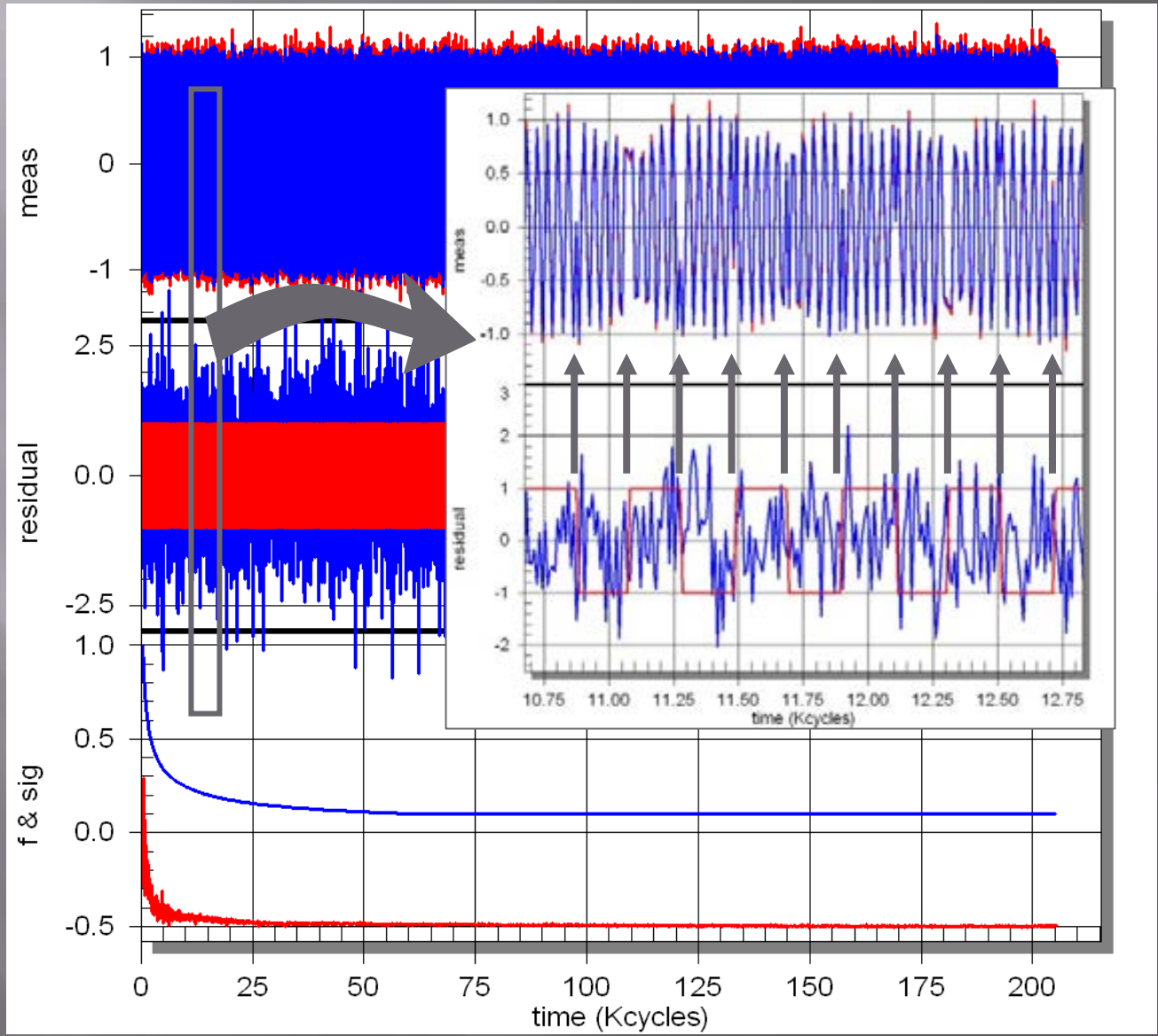


Normalized residual

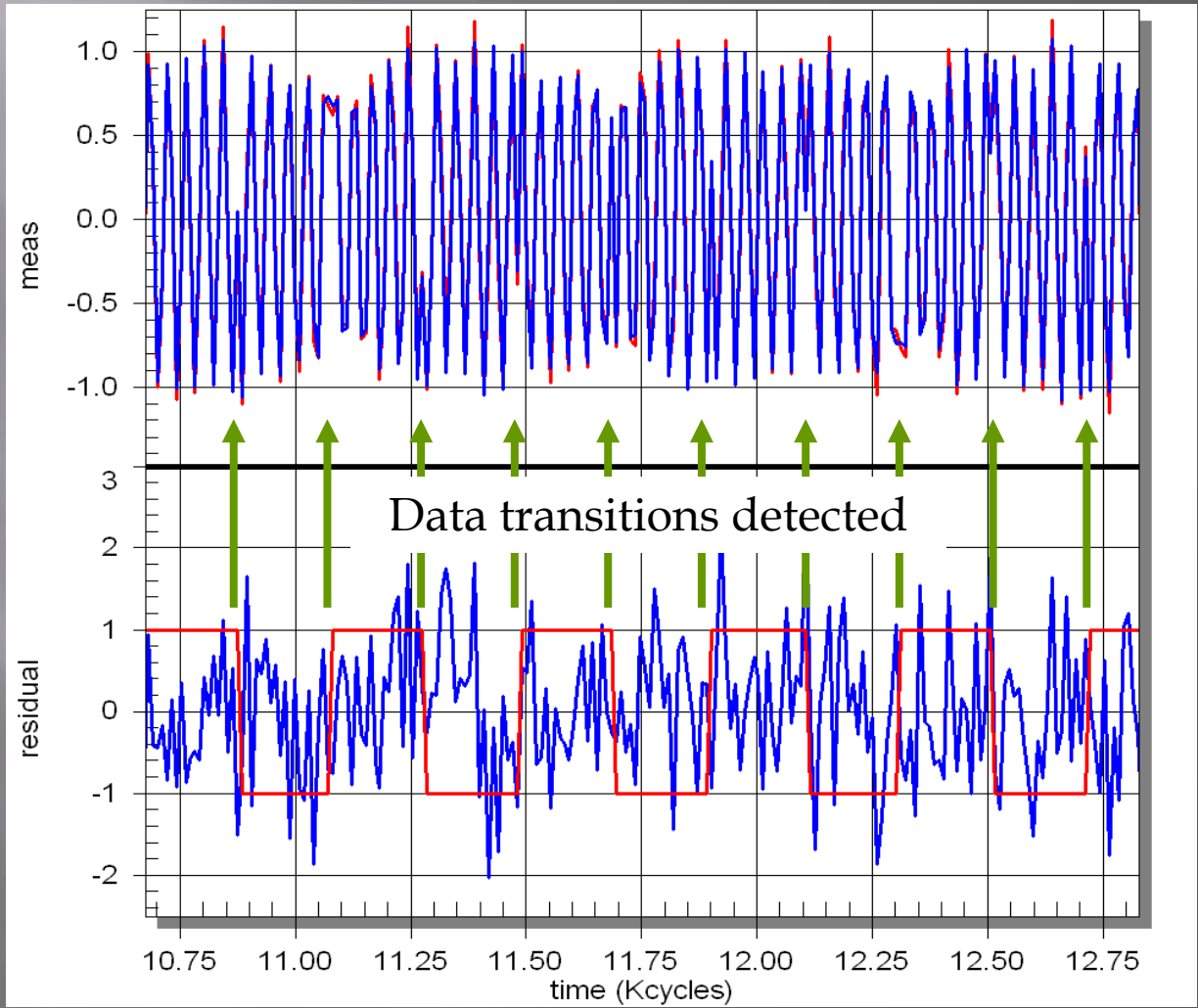


Two-class Hypothesis Test provides maximum Likelihood estimate of correct class

Representative Results



BPSK Data Transitions



Transition to Fixed Point

Use floating point implementation to define mantissa

- Replace all multiply & divide operations with a special procedure that tracks the maximum range of all internal variables
- Adjust scale factors so that range is +/- 2.0

Test Fixed point model

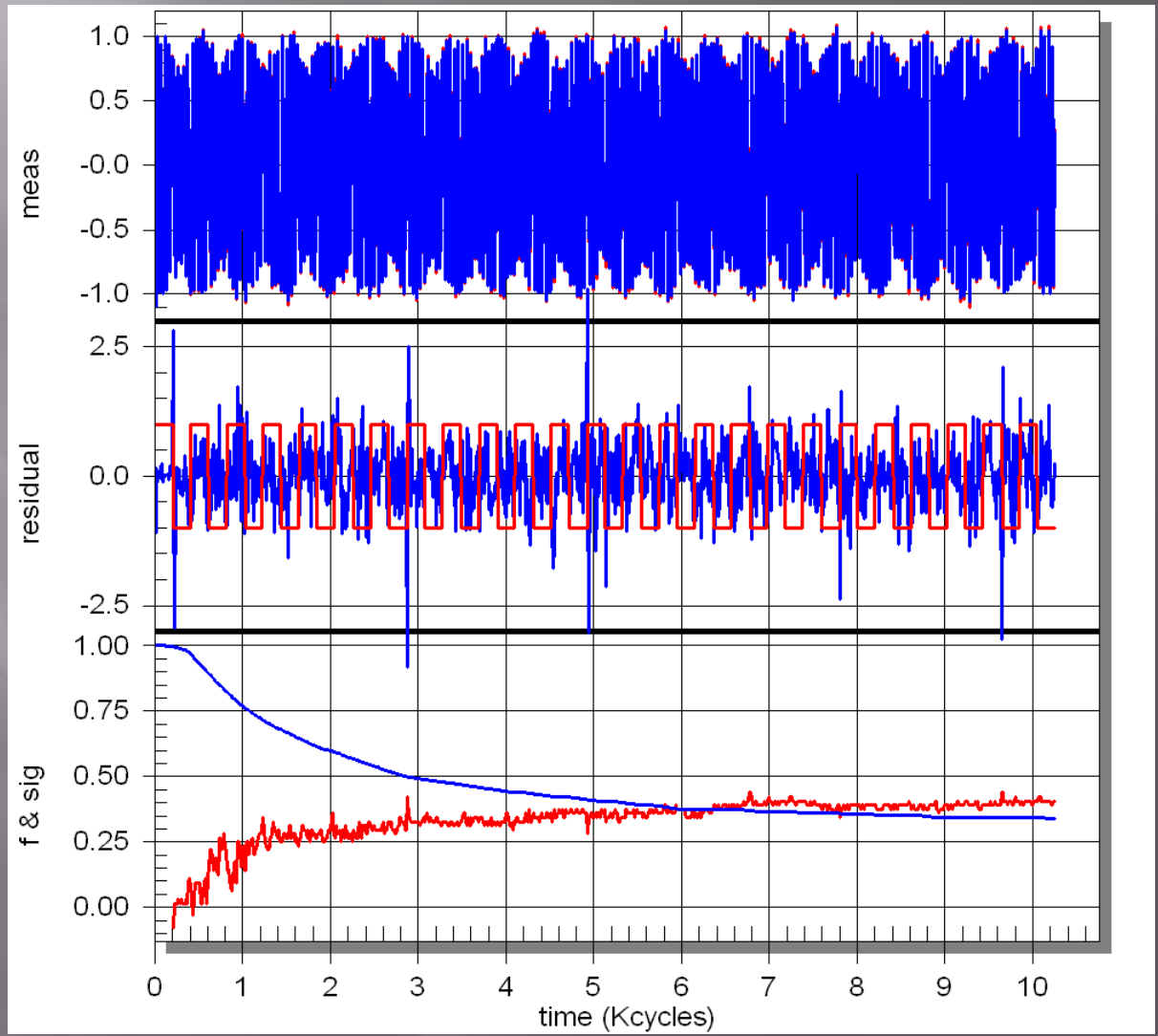
Replace all multiply and divide operations by
N = number of fraction bits

Multiply $$\left[C2^N\right] = ROUND\left(\left[A2^N\right]\left[B2^N\right]2^{-N}\right)$$

Divide $$\left[C2^N\right] = ROUND\left(\frac{\left[A2^N\right]2^N}{\left[B2^N\right]}\right)$$

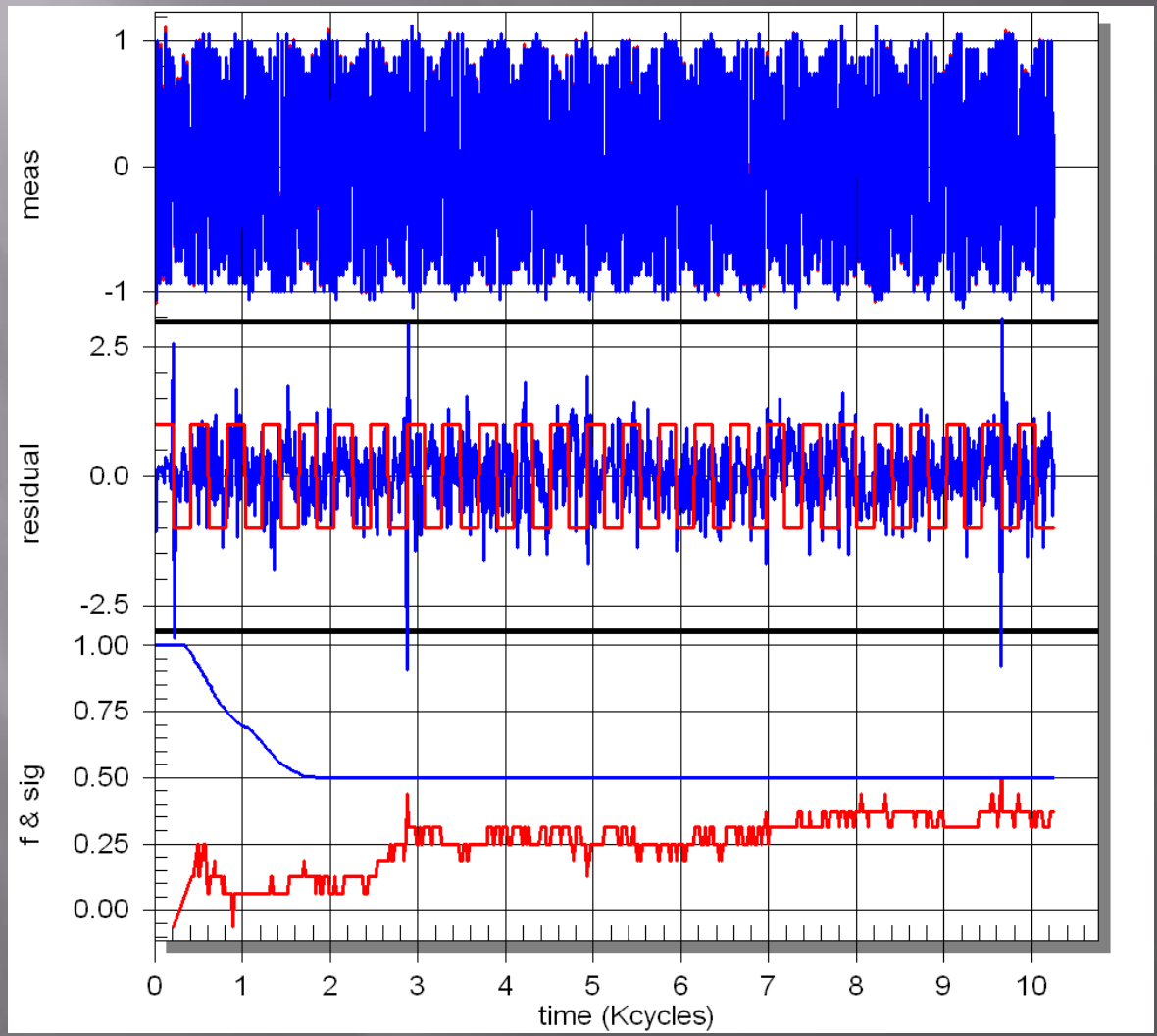
Integer SigmaRho Filter

12 fixed point bits 10 fraction bits



Integer SigmaRho Filter

10 fixed point bits 8 fraction bits



Next Directions …

100MHz sampling to decode data from carrier

- ✓ Direct carrier sampling
- ✓ De-spreading still required for initial CDMA lock & noise removal
- ✓ Replace frequency and code tracking with model-based KF
- ✓ Hypothesis testing for BPSK or QPSK or other encoding
- ✓ OFDM using multiple carrier models

SigmaRho fixed point filter makes VDLL practical

Leap forward SDR GPS + 3G/4G Comm

- ✓ Directly sampled carriers with software
- ✓ VDLL GPS
- ✓ Deeply embedded MEMS
- ✓ OFDM comm



GeoVantage Digital Imagery Systems

GeoVantage

Why IMU/GPS?

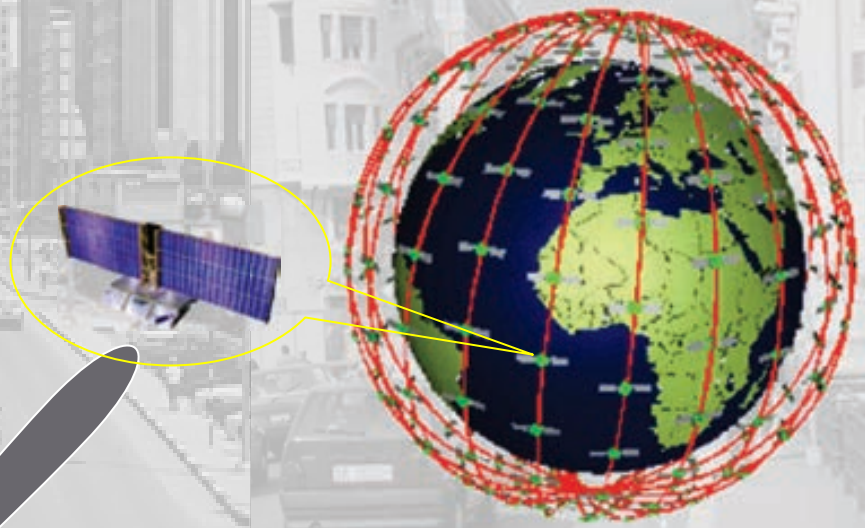
- **Make GPS Better**
  - ✓ **Feedforward to Tighten Track loops**
    - ❖ Densensitization to accel
    - ❖ Low C/N0 operation
    - ❖ Multipath rejection
  - ✓ **Smooth noisy position**
  - ✓ **Operate through GPS outages**
- **6DOF Vehicle Characterization**
- **Measure attitude**
  - ✓ **Antenna pointing**
  - ✓ **Remote Sensing**

Attitude Enables Anenna Pointing

Teledesic

Teledesic Enables Connected Car PC

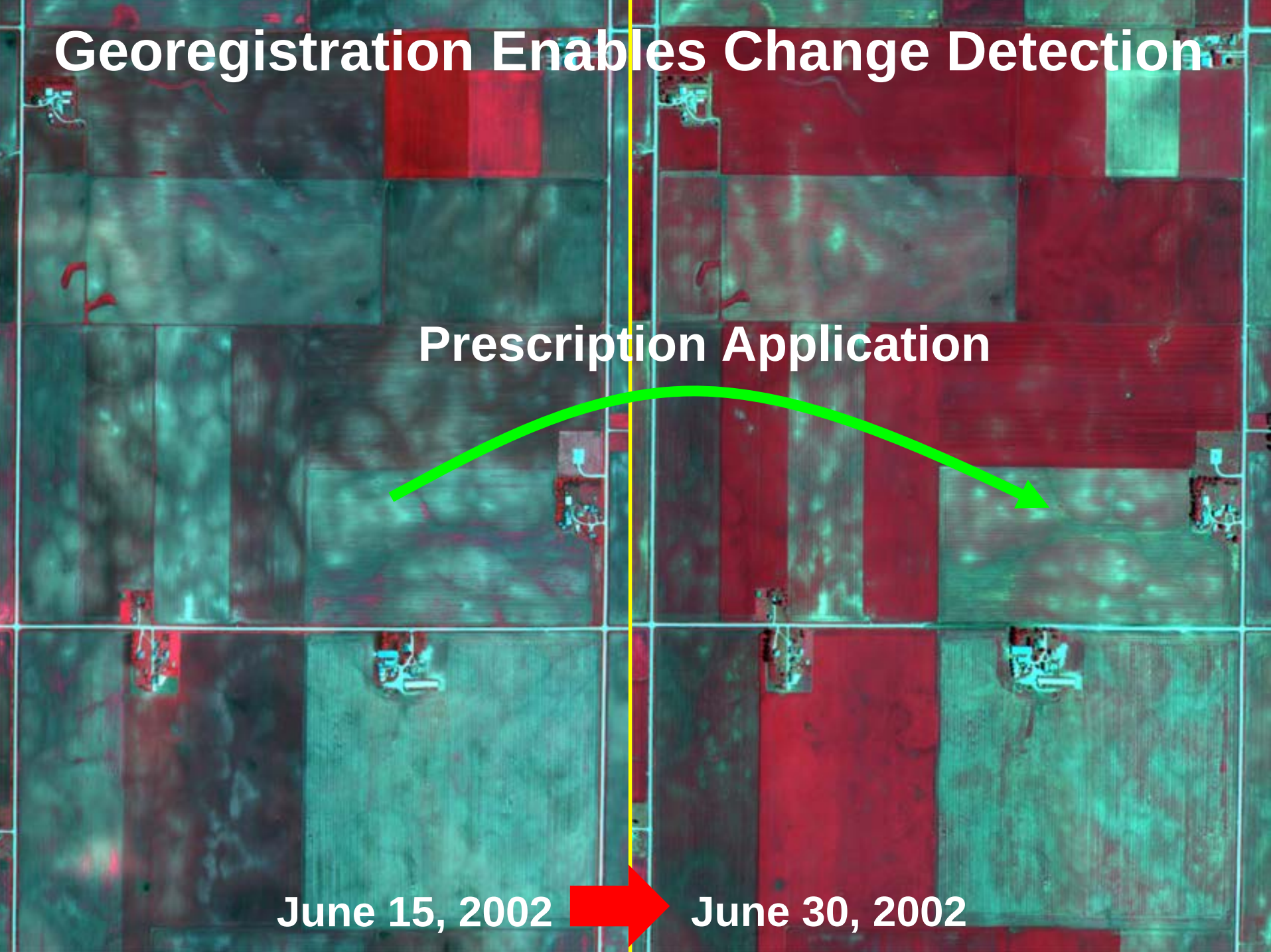




intel

connected car PC technology


Attitude Enables Georegistration


Georegistration Enables Change Detection
Prescription Application
June 15, 2002
June 30, 2002



International Operations

La Cumbre
Argentina







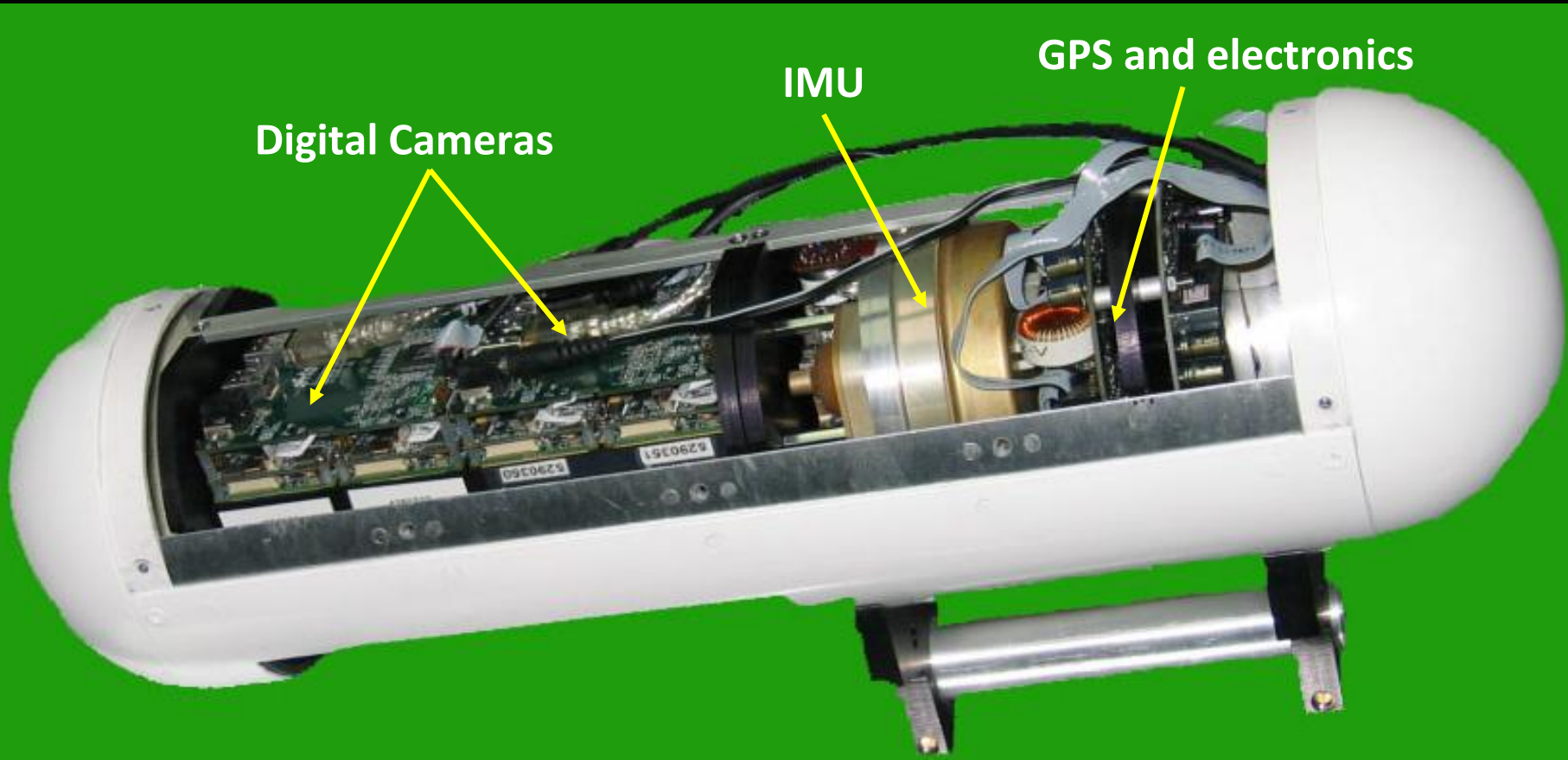
GeoVantage & Large Format simultaneous data collection

Flown	October 5, 2008
Resolution	Six inch





Geoscanner Digital Camera System

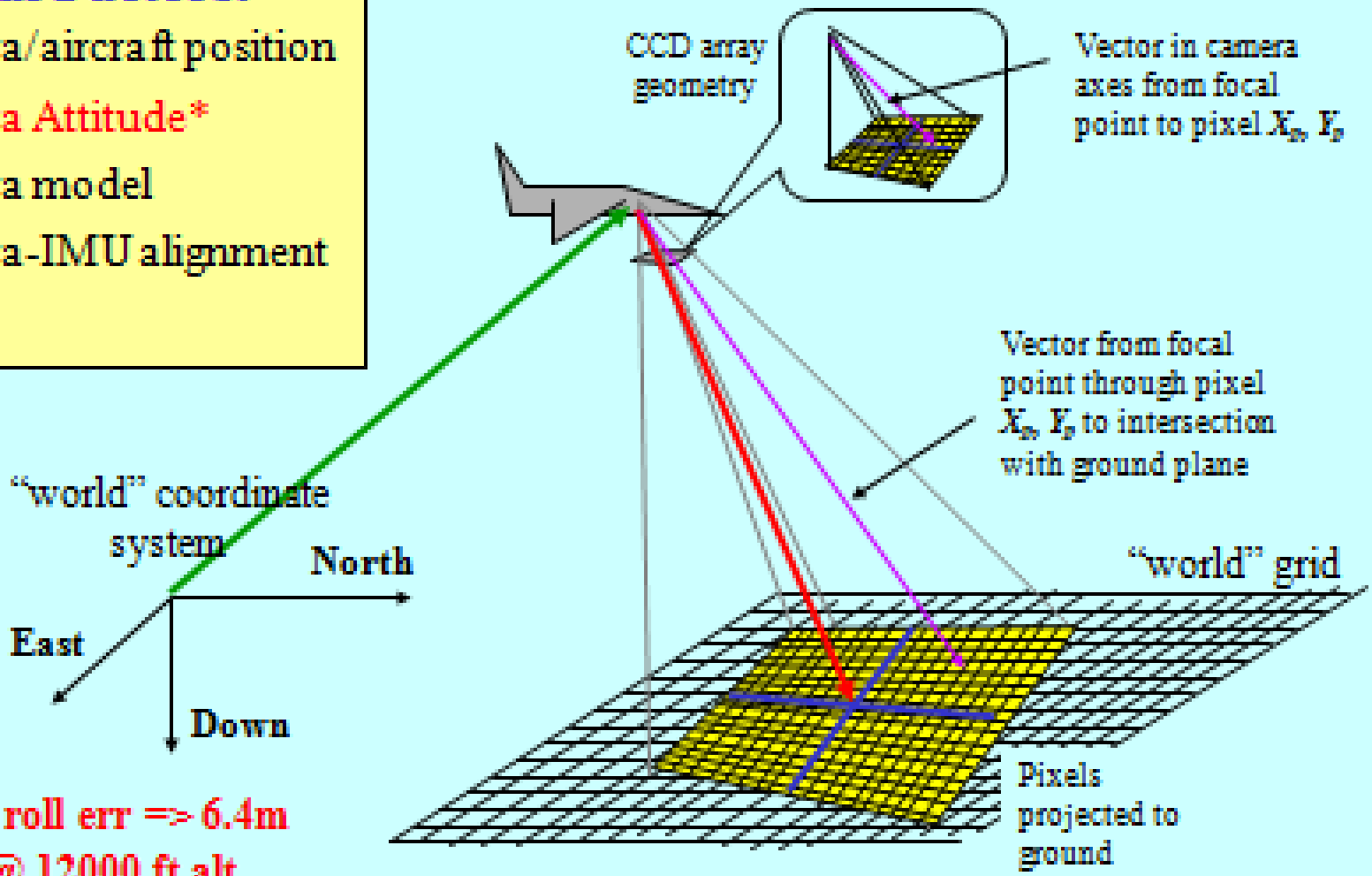


Single Frame GeoRegistration

WITHOUT Ground Control

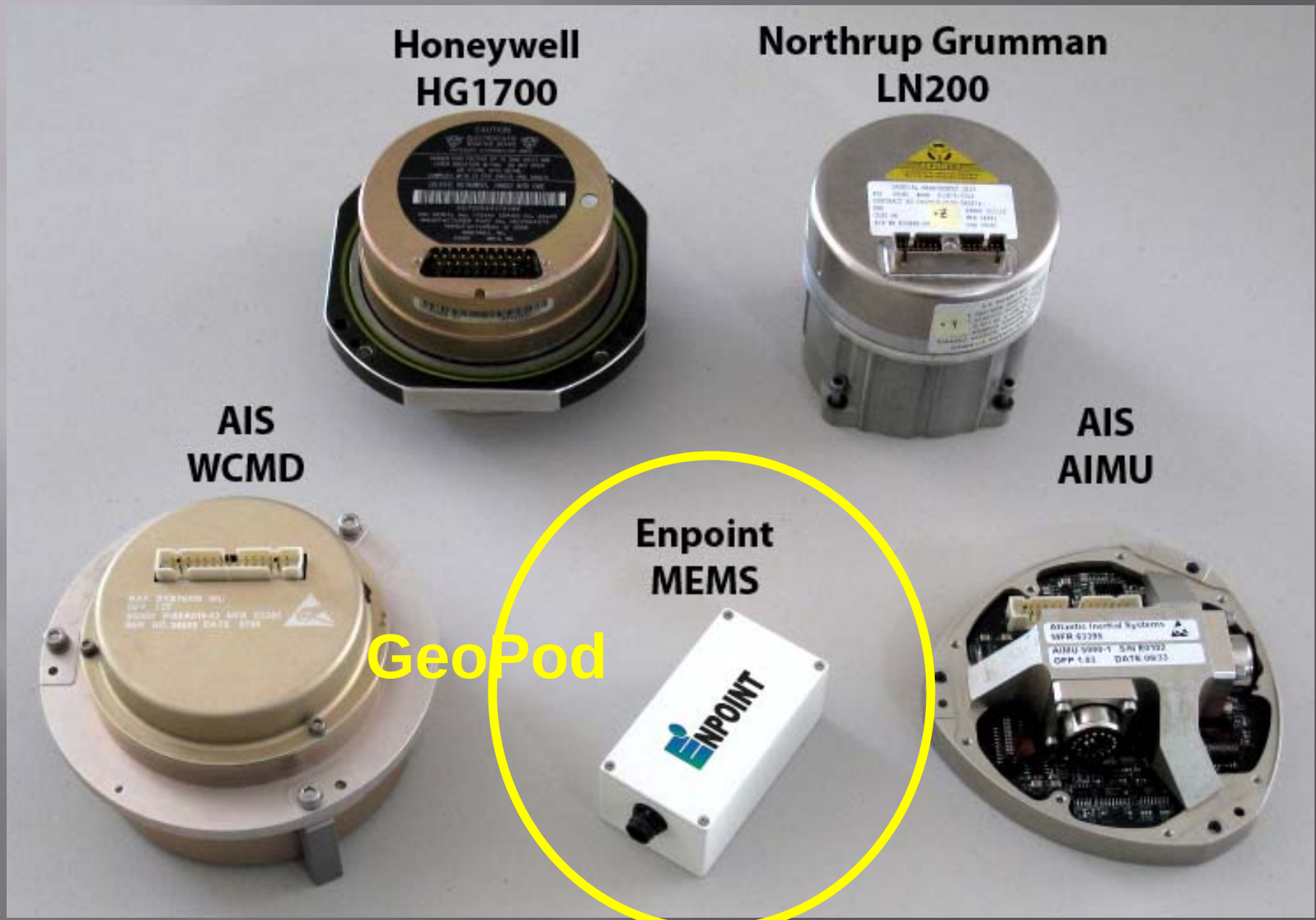
Critical Factors:

- Camera/aircraft position
- Camera Attitude*
- Camera model
- Camera-IMU alignment
- DEM



* 0.1 deg roll err => 6.4m error @ 12000 ft alt

Extensive IMU Integration Experience



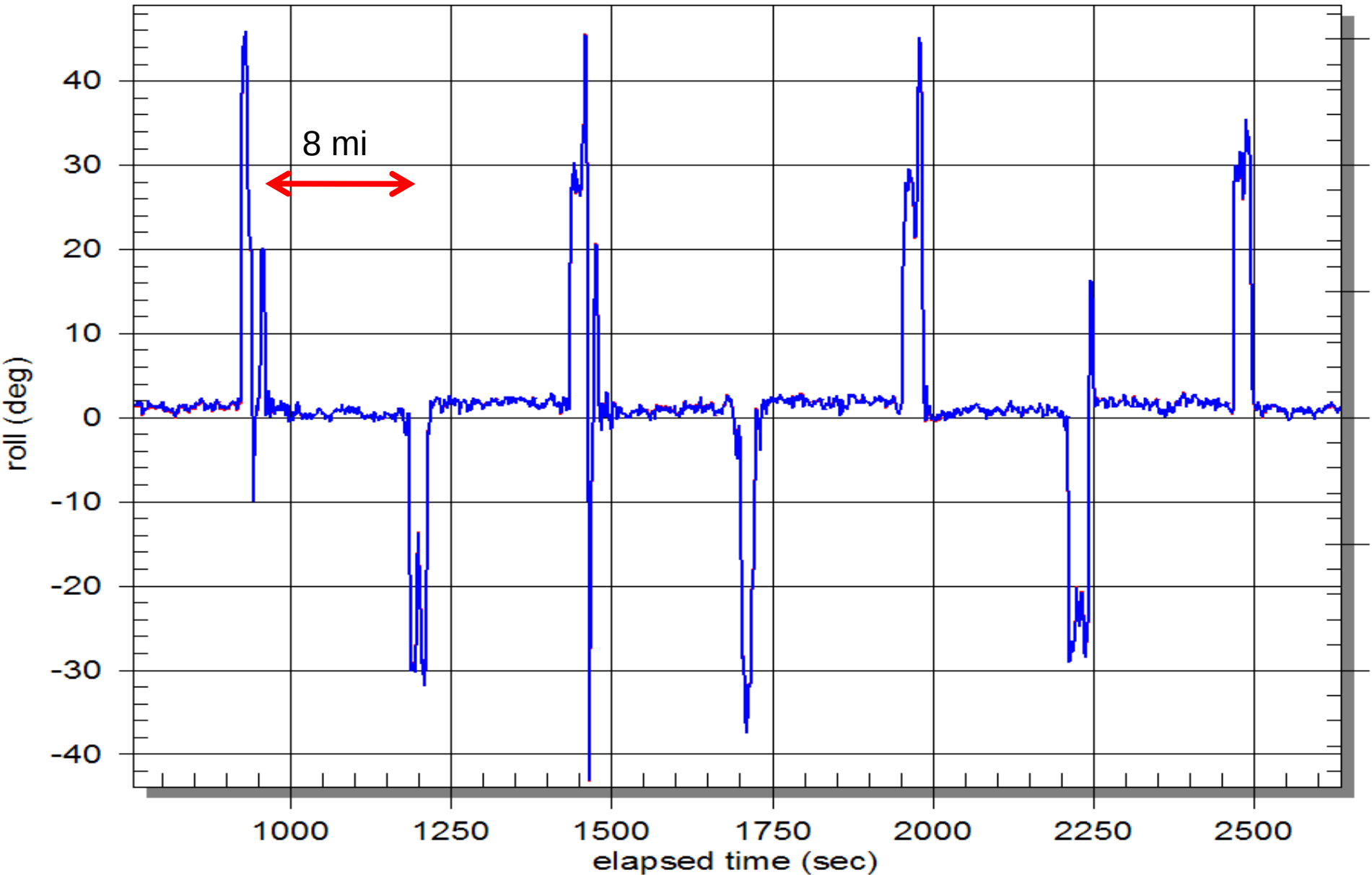
Flight Operations Improve Attitude Estimation

Technology and demand pushing imagery resolution to “inches” with equivalent georegistration accuracy

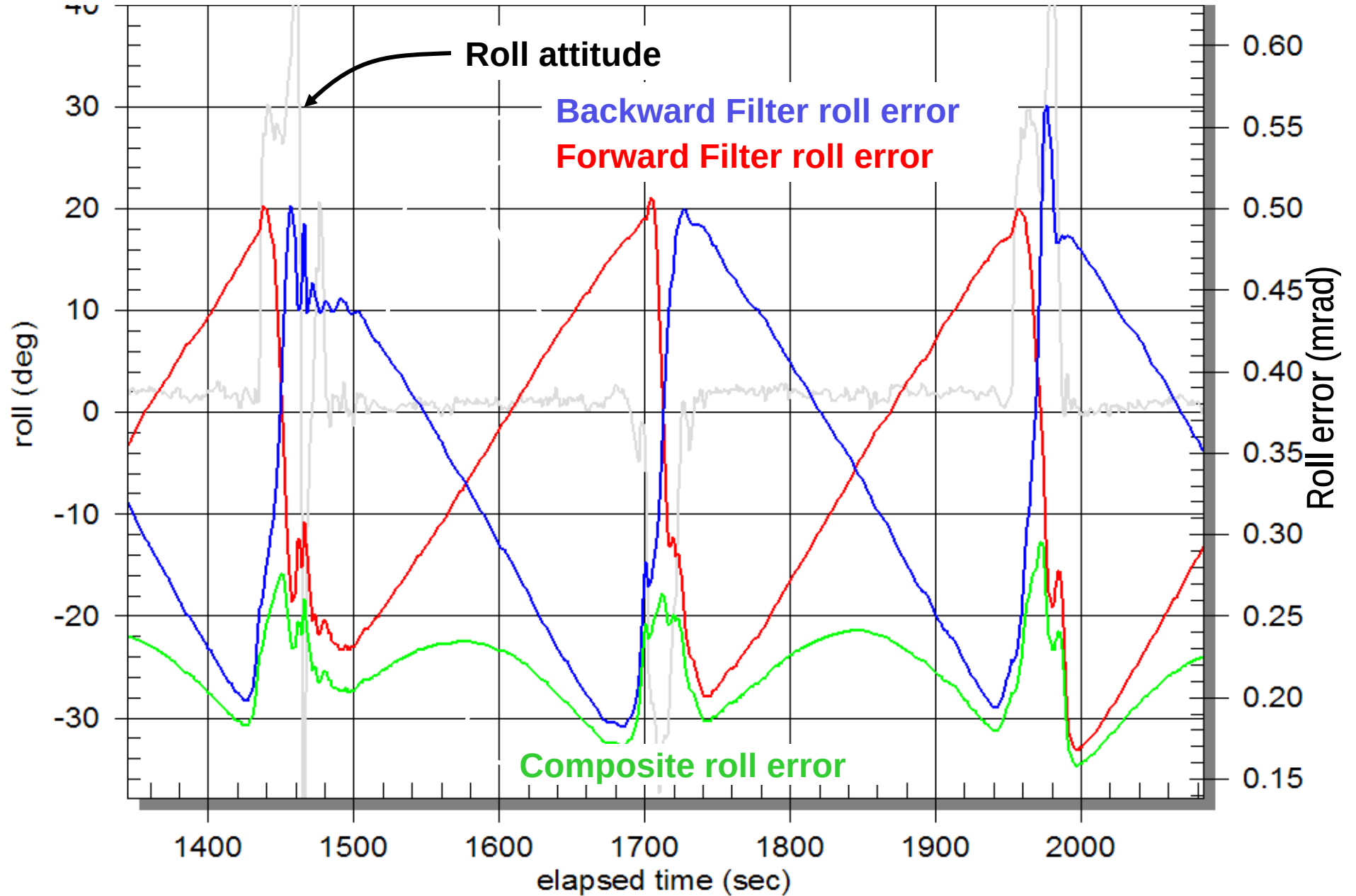
Can improve georegistration by:

- ✓ **Lower flight altitude**
- ✓ **Short flight lines**
- ✓ **Steep turns**

Representative Aircraft Dynamics



Kalman Filter Attitude Estimation



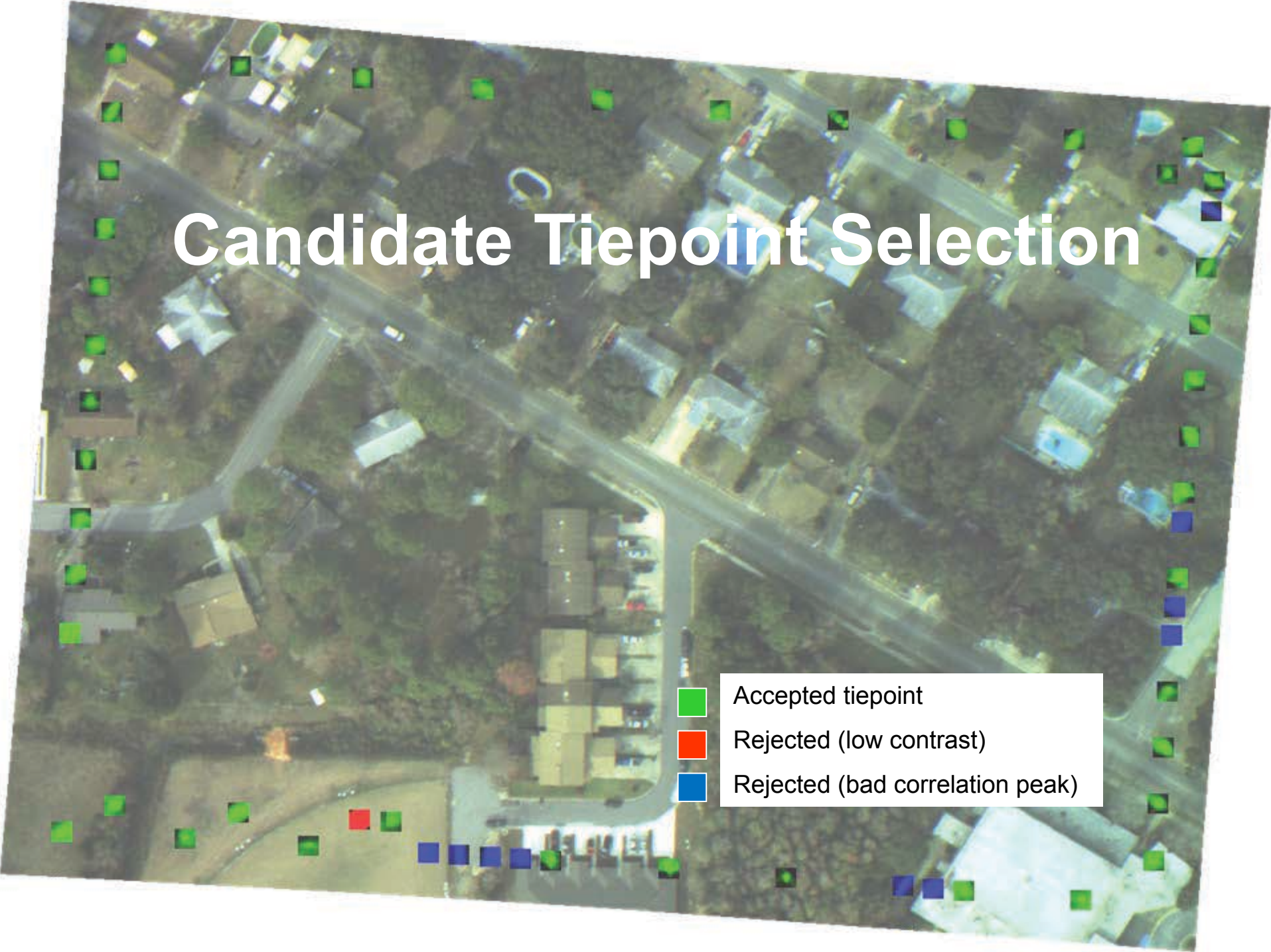
Seam Correction: Internal Tiepoints

Imagery with Seams is Unacceptable to clients

Solution:

- ✓ Place candidate tiepoints on single frames
- ✓ Locate match points on overlapping frames
- ✓ Adjust single frames for registration
- ✓ Use tiepoints for tonal balance

Candidate Tiepoint Selection




Candidate Tiepoints

Massive Use of Tiepoints



Complete automation:

40,000 matched tiepoints selected from
over 400,000 candidate tiepoints

IMU Specifications

Vendor and Model	Max Gyro Rate (deg/s)	Gyro In-run Bias (deg/hr)	Gyro SFE (ppm)	Angle Random Walk (deg/rt-hr)	Max Accel (g)	Accel In-run Bias (mg)	Accel SFE (ppm)	Velocity Random Walk (m/s/rt-hr)	Quantity Price ($)
Atlantic Inertial AIMU*	1400	2	350	0.05	40	0.3	700	0.03	11000
Honeywell HG1700*	1074	1	150	0.13	37	1.0	300	0.02	18500
Northrup Grumman LN200*	1000	1	100	0.07	40	0.3	300	0.08	27000
Analog Devices ADIS16365	300	25	1000	1.90	17	0.2	3000	0.35	300

* ITAR control required

Access to Commercial MEMS Navigation



Flash Boot Prom and File System

2 GPIO

Memory

4 GPIO



ADI 21469 SHARC

SPORT, Linkport

Multiple GPIO

UART

SPI

Memory Bus

SPI Timer

SPORT, PPI or Host DMA

ADI BF5xx Blackfin

UART

UART

Timer/GPIO

? UART USB SPORT

RS422 — Synchronization Source

RS232 — Bidirectional Communications

RS232 — GPS Receiver



RS422 — Synchronization Sink for AIMU

? — Device Control

RS232 — Motor Control

? — AIMU Communications

? — Bidirectional Communications

RS232 — GPS Receiver

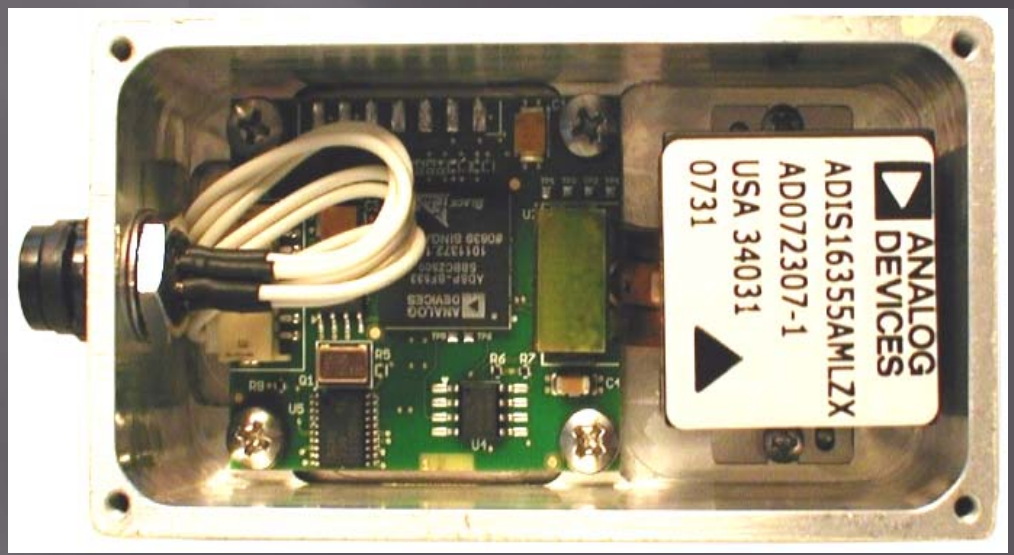


MAC/PHY — LAN

MEMS IMU Hardware Features

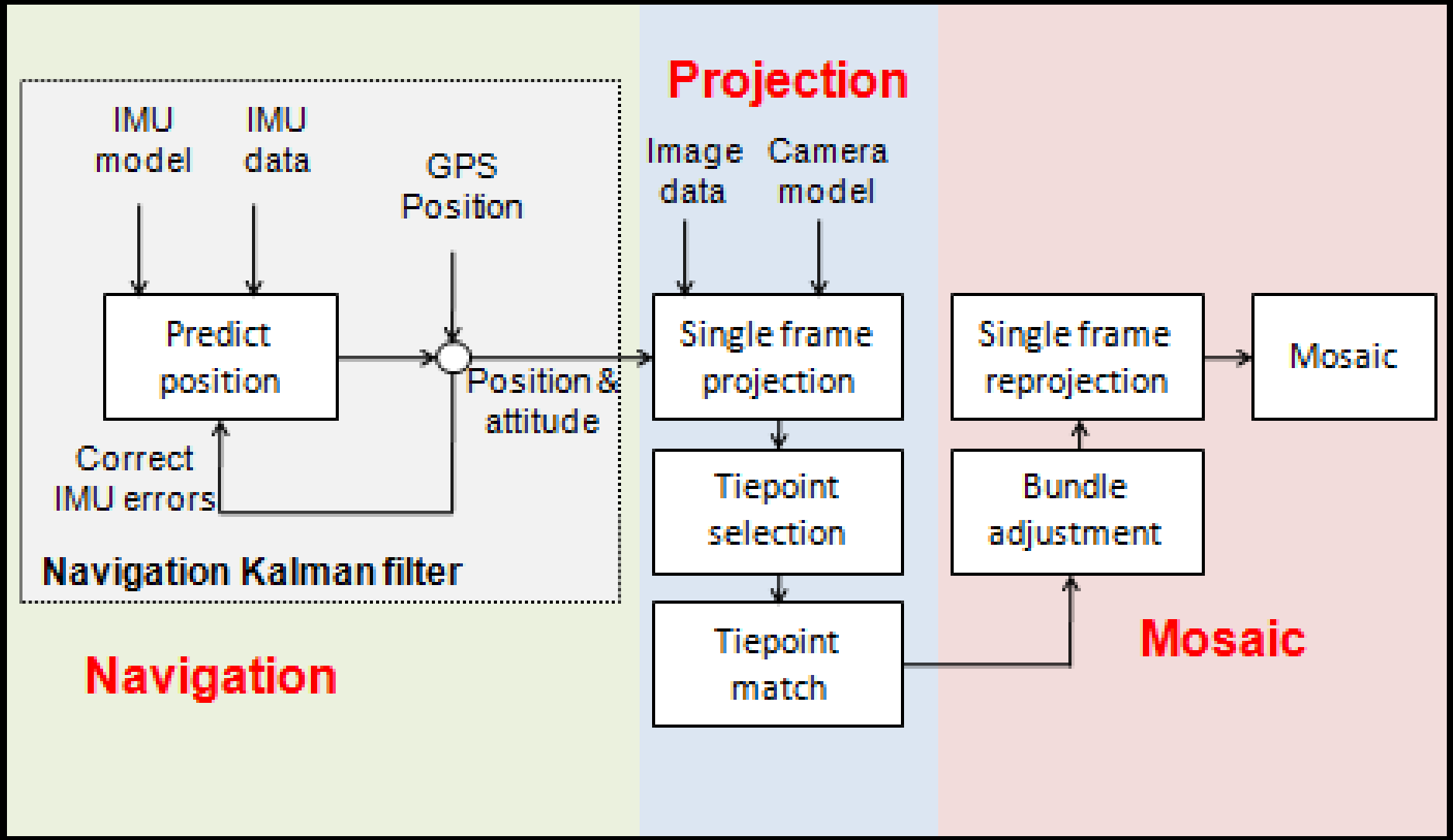
- ADI ADIS1635x
  - 3-axes acceleration measurement
  - 3-axes attitude rate measurement
- ADI Blackfin BF533 or BF538 for application support
- 3-axis magnetometer
- SDRAM – up to 32MB
- Flash – up to 2MB
- Protection: over voltage, ESD, polarity



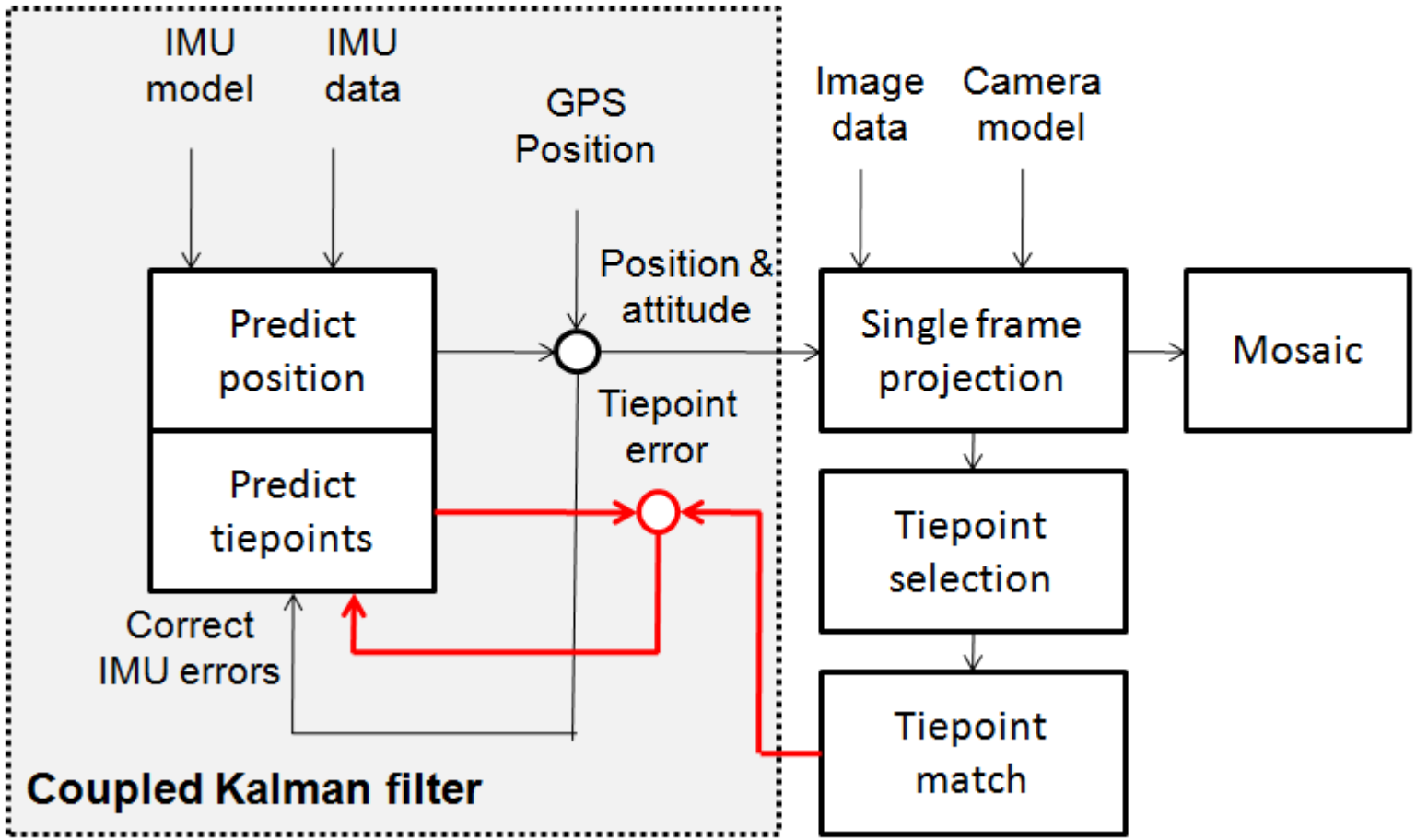


Traditional Tiepoint Mechanization

Three sequential steps



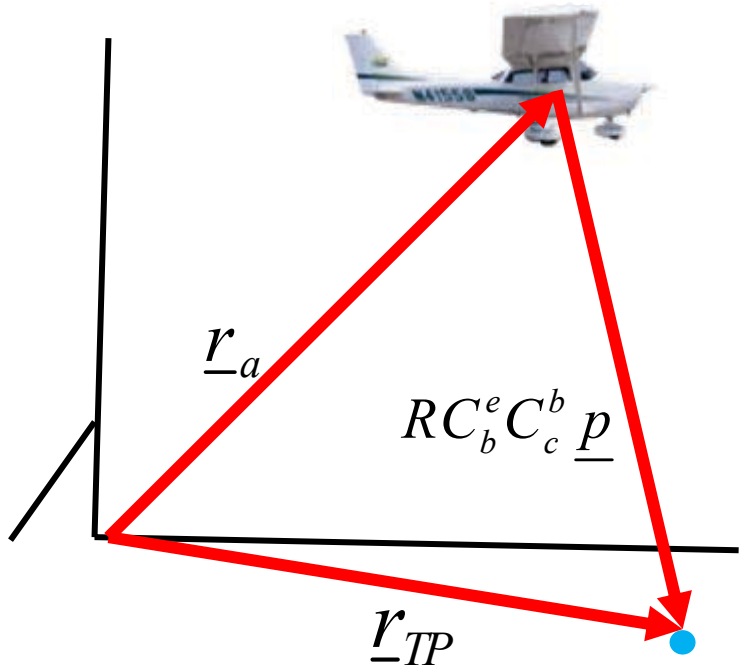
Imagery Aided Navigation



Navigation Aiding from Mosaicked Imagery

$$\underline{p} = \begin{bmatrix} fl \\ \varepsilon Y_{pix} \\ \varepsilon Z_{pix} \end{bmatrix} / \sqrt{fl^2 + \left(\varepsilon Y_{pix}\right)^2 + \left(\varepsilon Z_{pix}\right)^2}$$

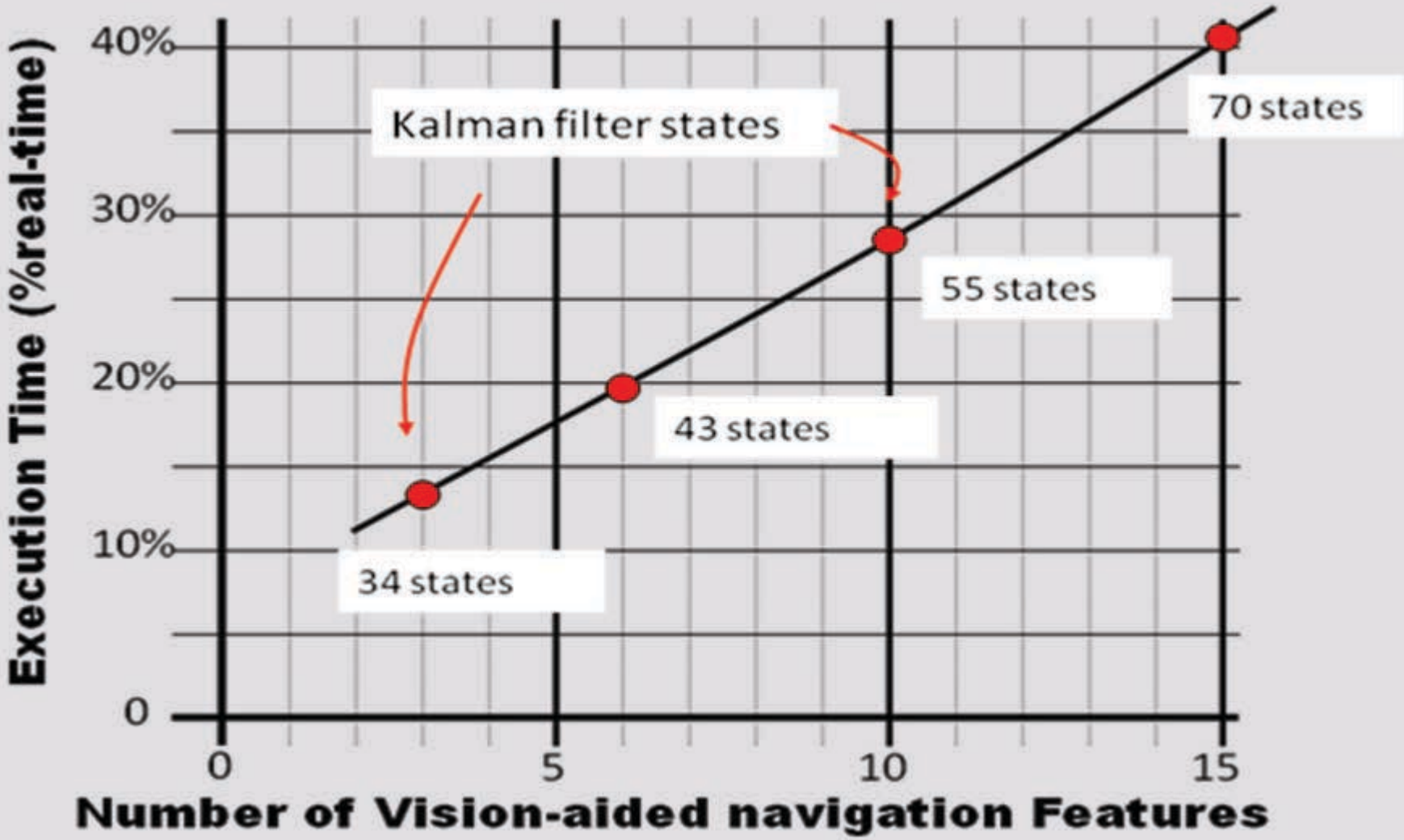
$$R = \frac{h_a - h_g}{\left[C_b^e(t_1)\underline{p}_1\right] \bullet \underline{\hat{g}}}$$



$$\begin{bmatrix} \dot{Y}_{pix} \\ \dot{Z}_{pix} \end{bmatrix} = fl' \left[\begin{array}{c|c|c} \frac{p_2}{p_1} & \frac{1}{p_1} & 0 \\ \hline \frac{p_3}{p_1} & 0 & \frac{1}{p_1} \end{array}\right] \left[C_e^b \frac{\dot{\underline{r}}_a}{R} + \frac{\left(\underline{r}_a - \underline{r}_{TP}\right)}{R} \bullet \frac{\dot{\underline{r}}_a}{R} \underline{p} + \underline{\omega}_{b/e}^b \times \underline{p} \right]$$

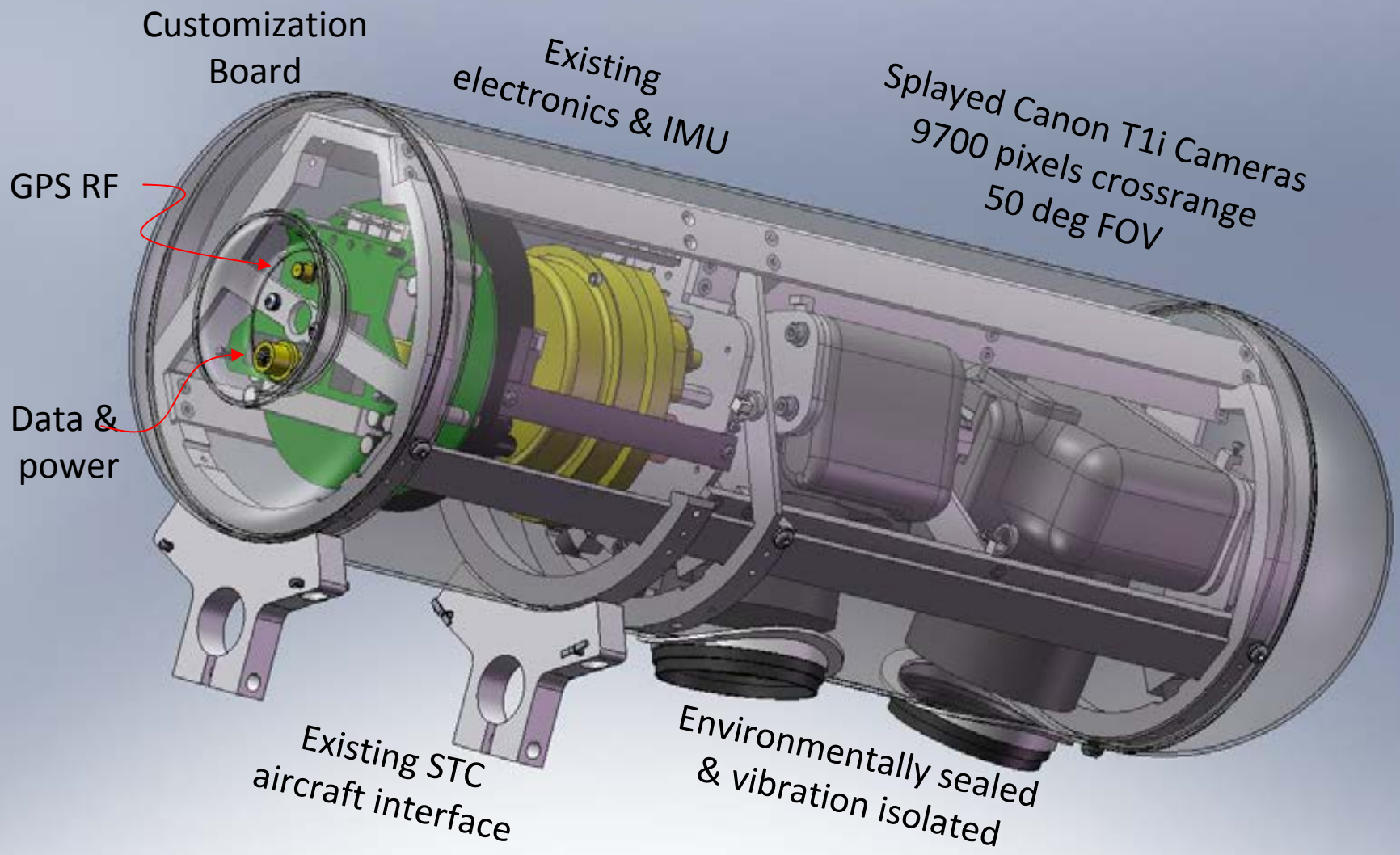
Propagate tiepoint to next observance

SigmaRho Efficiency for Large State Vectors



GeoVantage GeoPod Design

Goal: 10000 pixel crossrange, non-ITAR



Web Information Flow



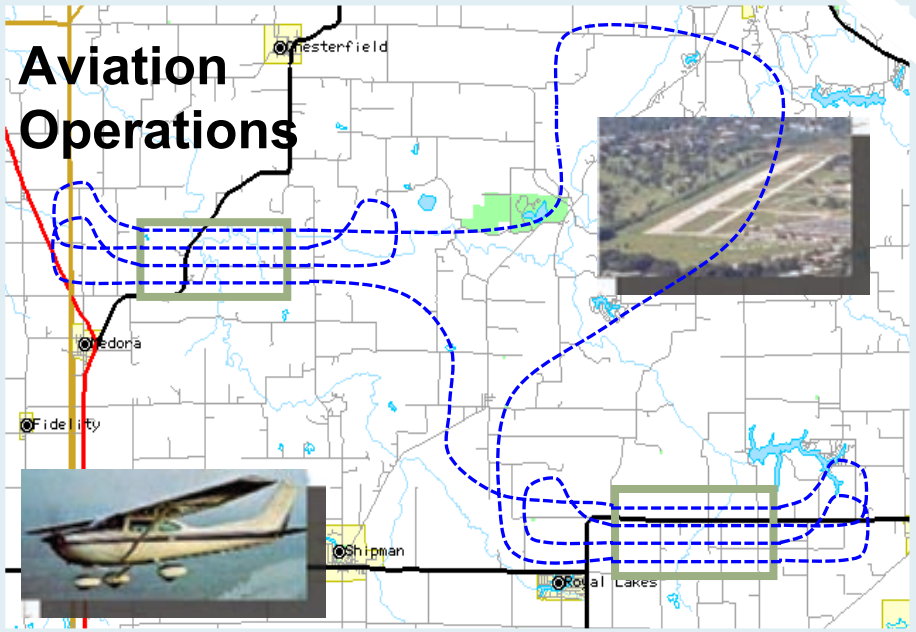
- Data Request
- Data status
- User Help
- Feedback



Internet

- Mission status
- Hardware status
- QC metrics

Imagery Products



- Mission Plans
- Processing commands



Aircraft Flight Computer

GeoVantage Business Goals

- Scale to 1000 RS systems
- Heavy International Focus
- Low Price/area (below satellite)
- Fast Response Times (<one day)
- Web Order-to-Delivery with web status
- Standardized RS Products
- Automate Everything